

Fundamental nuclear physics at the exascale and beyond

Robert Edwards
Jefferson Lab

Fundamental nuclear physics at the exascale and beyond

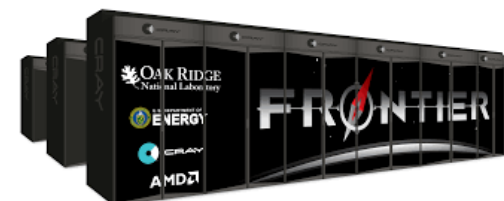
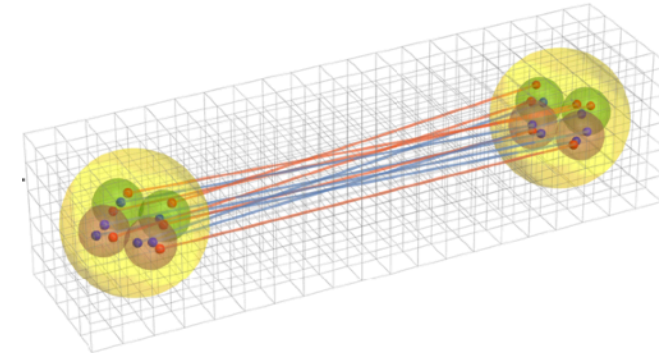
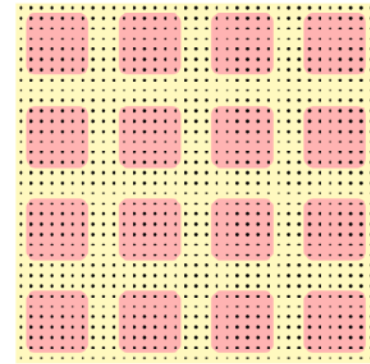
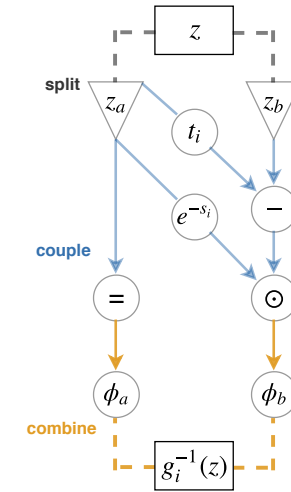
LQCD ASCR/NP SciDAC-5

- Argonne:
 - Robert Latham
 - Sandeep Madireddy
 - Yong Zhao
- Brookhaven:
 - Swagato Mukherjee
- JLab
 - Jie Chen
 - Robert Edwards
 - Joseph Karpie
 - Frank Winter
- Lawrence Berkeley
 - Aydin Buluc
 - Sherry Li
 - Andre Walker-Loud
- Los Alamos
 - Tanmoy Bhattacharya
- MIT:
 - Saman Amarasinghe
 - Will Detmold
 - Andrew Pochinsky
 - Phiala Shanahan
- Oak Ridge:
 - Prasanna Balaprakash
 - Henry Monge Camacho
- William & Mary:
 - Kostas Orginos
 - Andreas Stathopoulos
- NVIDIA:
 - Kate Clark
 - Balint Joo

SciDAC-5 partnership - ASCR & NP

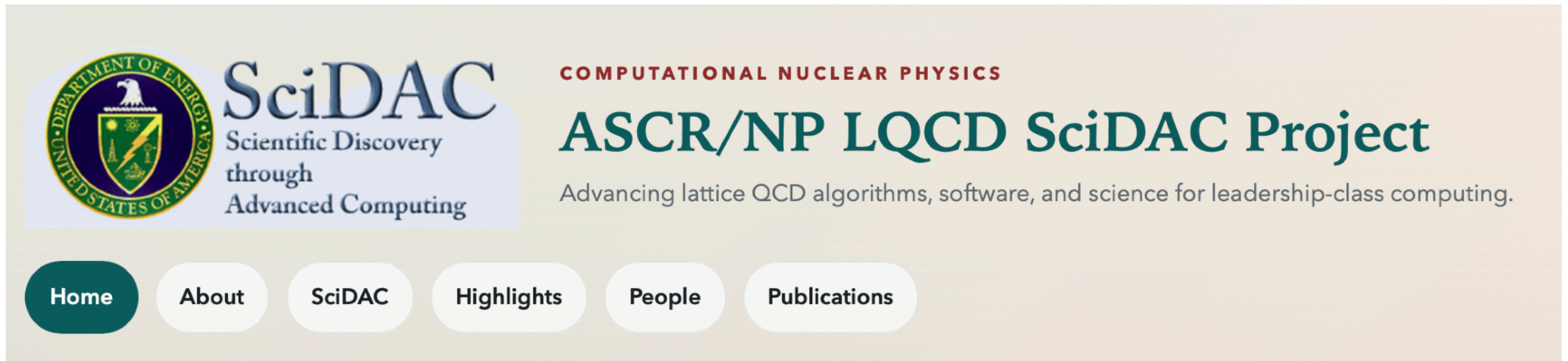
Three main themes:

- Gauge-field generation
 - *Use ML and hierarchically structured sampling*
 - Impact - radically increase fidelity and statistics of calculations
 - Partnership with FastMath
- Analysis
 - *Linear system solvers and many-body correlation function computations*
 - Impact - extract ever more sophisticated physics processes
 - Partnership with RAPIDS & FastMath
- Changing computational landscape
 - *Code-portability and development for new hardware architectures*
 - Impact - utilize advanced capabilities of Frontier, **Aurora**, Doudna & Discovery
 - Partnerships with RAPIDS & FastMath



Publicity and useful repositories

Website: <https://lqcdscidac.github.io>



Under SciDAC-5

- > 60 Journal articles
- > 15 Letters

Gratuitous plug!

<https://usqcd.github.io>

USQCD: All-years totals:

Publications: 4513

Citations: 205960

arXiv search: author $\in$ USQCD_members AND category=hep-lat.

- **Fundamental Discovery:**

- Advancing first-principles calculations of hadron spectroscopy, hadron structure, and nuclear structure to reveal how the strong force builds visible matter and to provide new pathways for discovering physics beyond the Standard Model.

- **Science and Technology Innovation:**

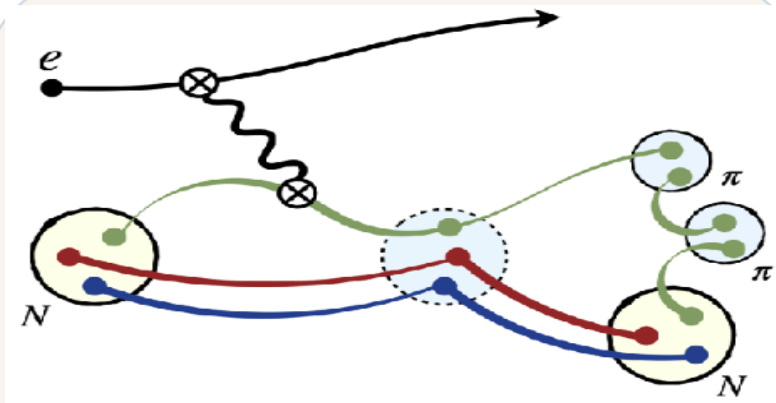
- Developing scalable algorithms, portable software, AI-enabled analysis, and integrated workflows that turn leadership-class computing into sustained scientific capability.

- **National Scientific Capability:**

- Providing essential theory support for major experimental programs, extending the scientific reach of DOE computing facilities, and training the next generation of computational nuclear scientists.

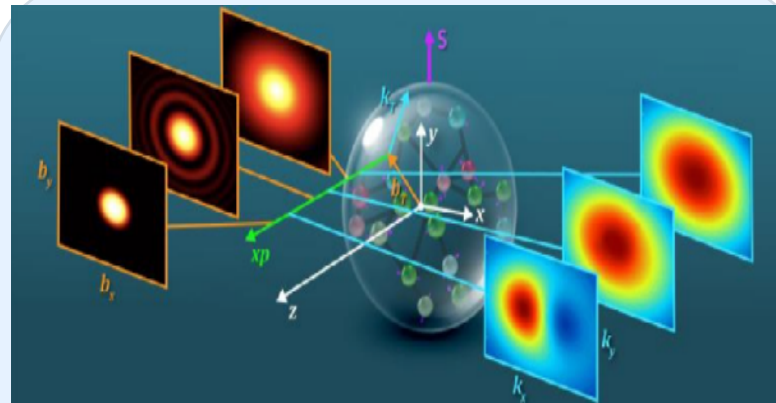
Science drivers

QCD calculations connect experimental programs to the quark & gluon dynamics of matter



Hadron spectroscopy

Predicting dynamics of particles and resonances



3D femtoscale imaging

Understanding the quark/gluon structure of matter from QCD



Nuclear structure

Precision nuclear & neutrino observables

Spectroscopy

Femtосcale imaging

Nuclear interactions

Precision / BSM

Connection to Experimental Programs

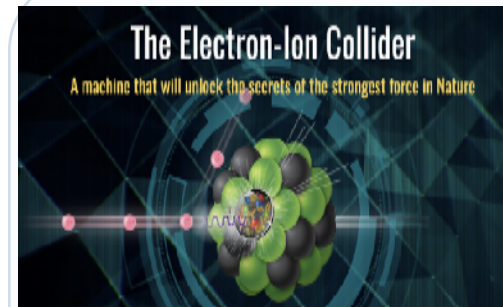
Maximizing discovery potential of next-generation experiments require QCD input



Jefferson Lab
Hadron spectroscopy
& structure



DUNE / LBNF
Neutrino interactions



Electron-Ion Collider
3D nucleon & nuclei
structure



HL-LHC
High-luminosity
frontier

Precision inputs

QCD results for nucleon / nuclear properties are **key inputs** for experimental success

Data movement

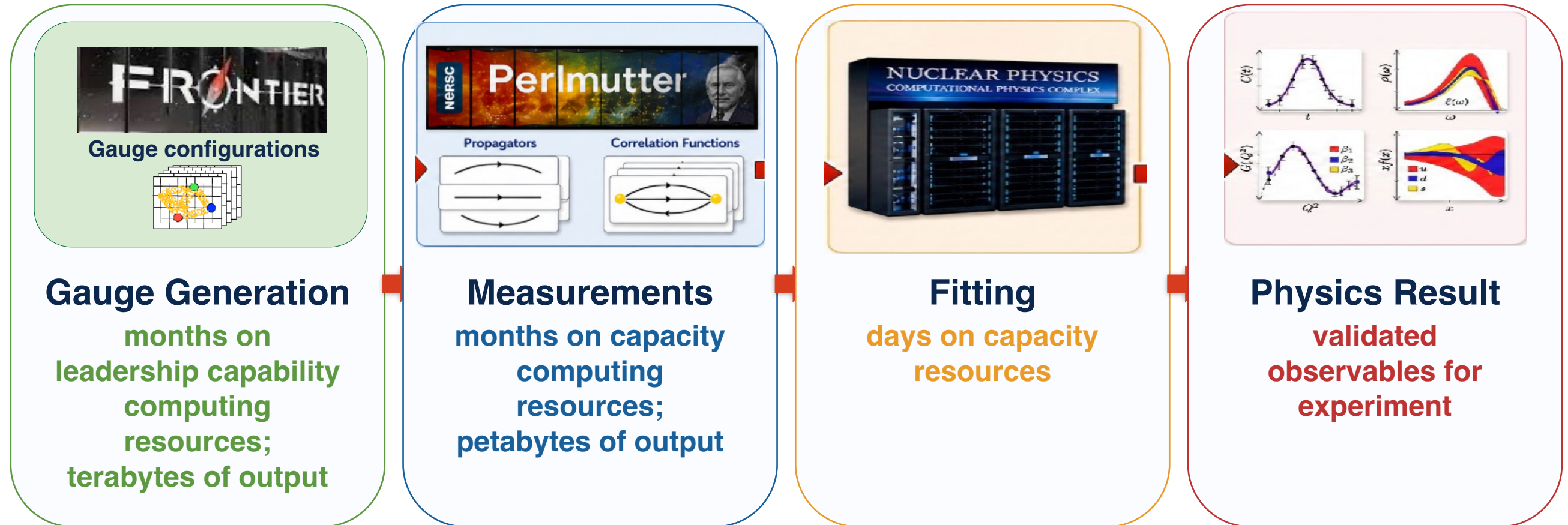
QCD data is large (many PBs) and **reusable** for multiple applications

Automation

QCD workflows are complex opportunities for automation

Challenges

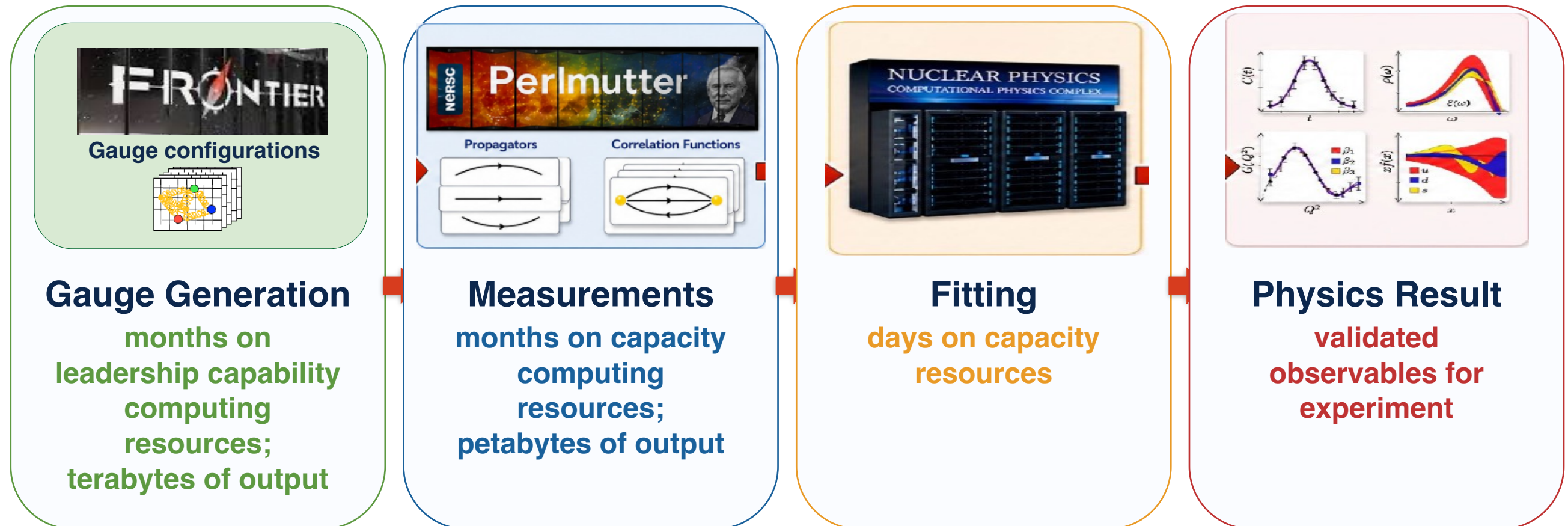
Workflows span capability machines, capacity clusters, & user-intensive analysis steps



SciDAC 1-5 has been crucial at advancing all stages of the pipeline

Challenges

Workflows span capability machines, capacity clusters, & user-intensive analysis steps



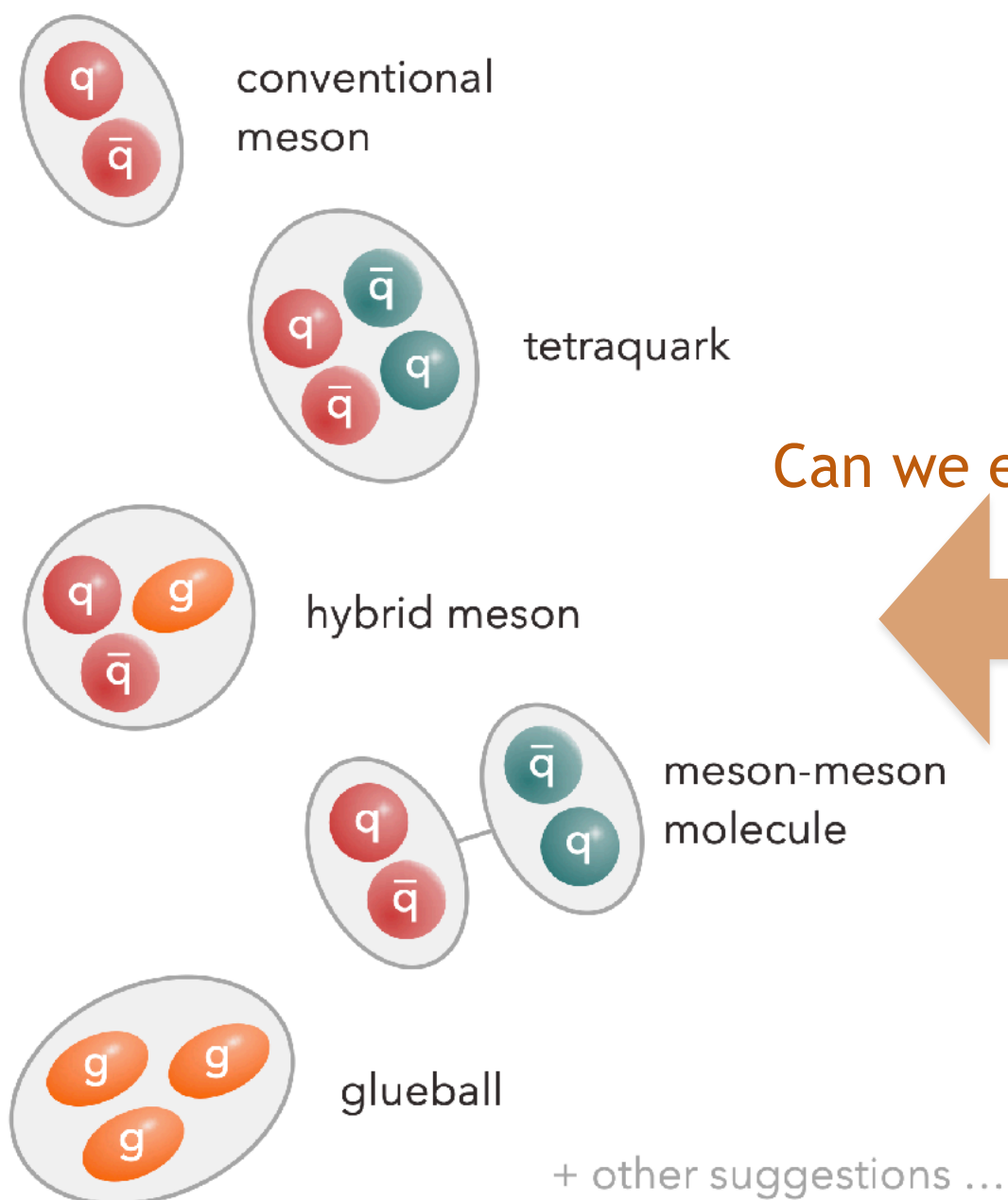
SciDAC 1-5 has been crucial at advancing all stages of the pipeline

Suitable for AI Agentic controls

Genesis Mission | AMERICAN SCIENCE CLOUD

(Meson) spectroscopy

“pictures”

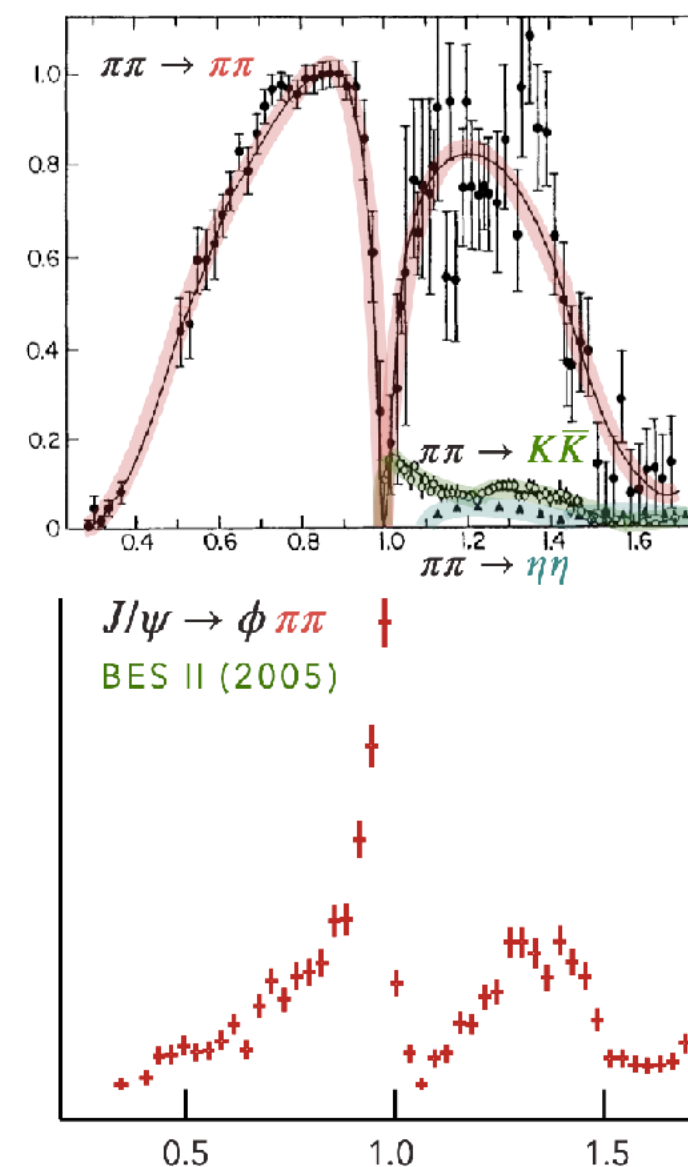


Realm of modeling
connection to QCD unclear

e.g. $f_0(980)$

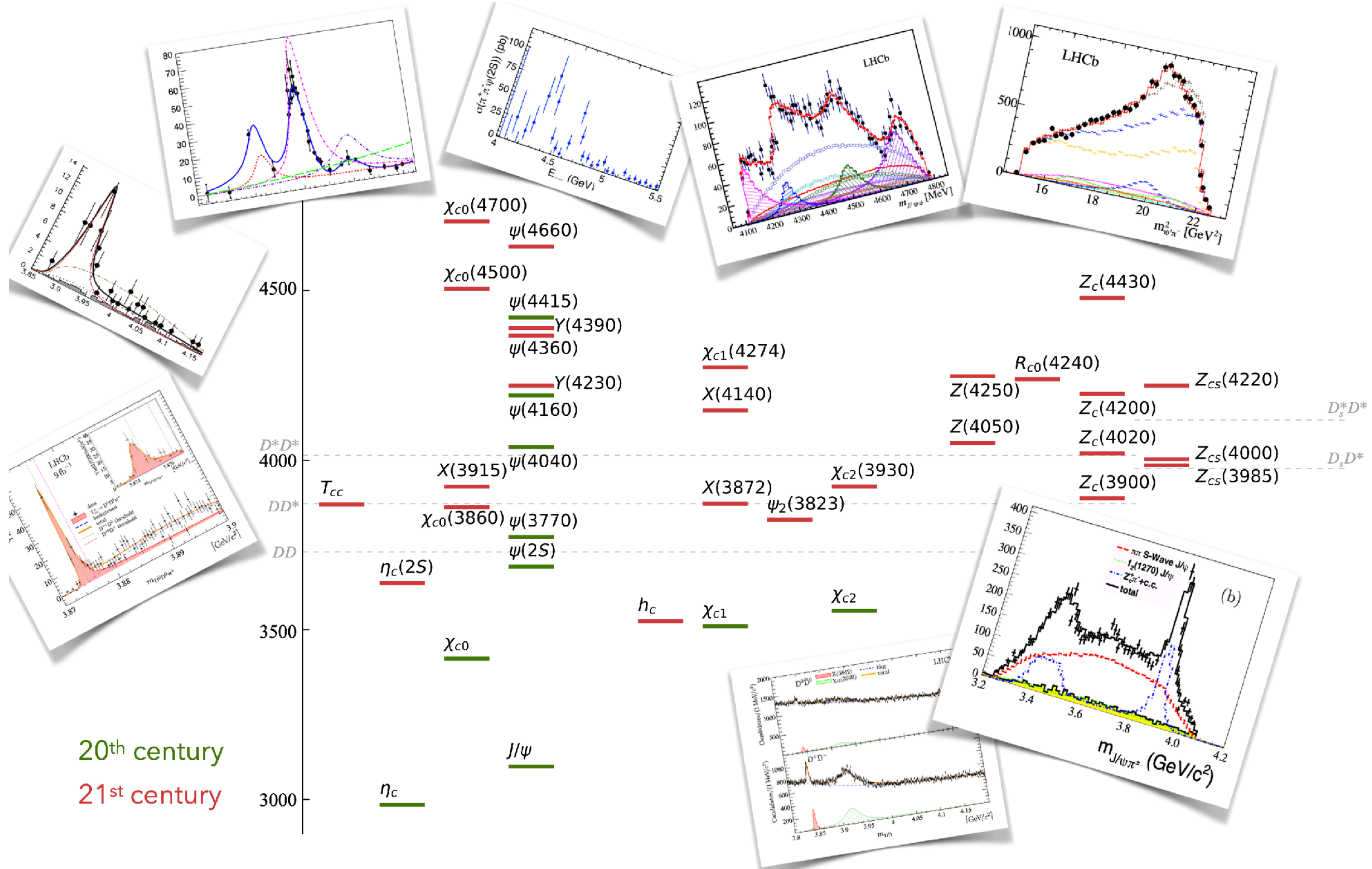
Can we establish connection?

“reality”



Measurements of stable hadrons
connection to QCD unclear

Renewed motivation - the XYZ explosion - tetraquarks?



Charmonium resonances from coupled-channel QCD

First comprehensive calculation:

First LQCD determination of scalar and tensor charmonium resonances with all relevant coupled two-meson channels included.

A surprisingly simple spectrum

One scalar & one tensor resonance; no proliferation of extra states despite the many experimentally observed XYZ candidates.

Enabled by SciDAC

Automated graph evaluation developed with the SciDAC RAPIDS partnership, together with scalable correlation-function workflows, made the coupled-channel calculation possible.

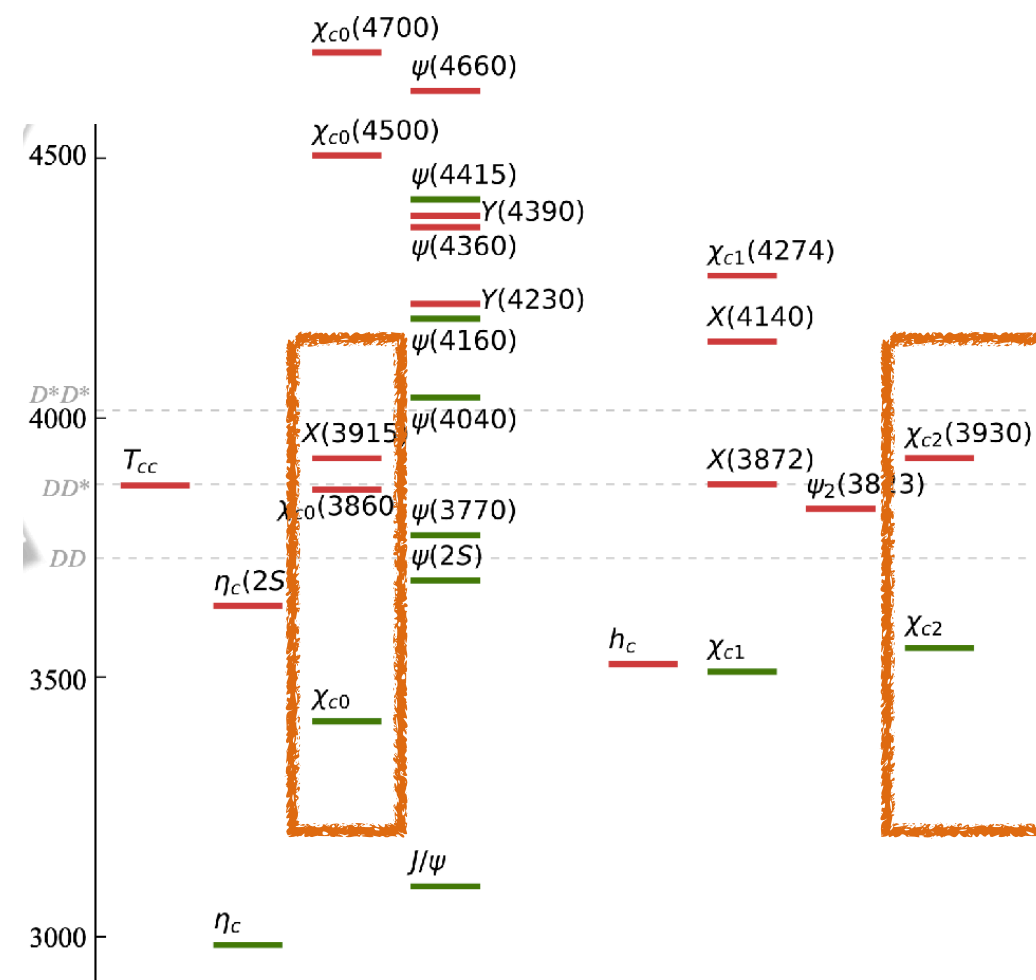
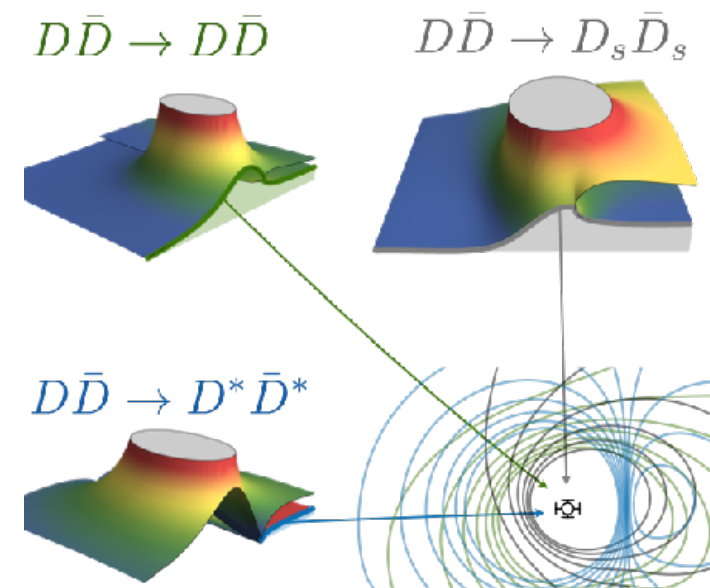
Experimental impact:

Charmonium spectroscopy is an intense focus at experiments worldwide.

The experimental spectrum is crowded; the underlying QCD resonance content may be considerably simpler.

DOE highlight; Editor's Suggestion

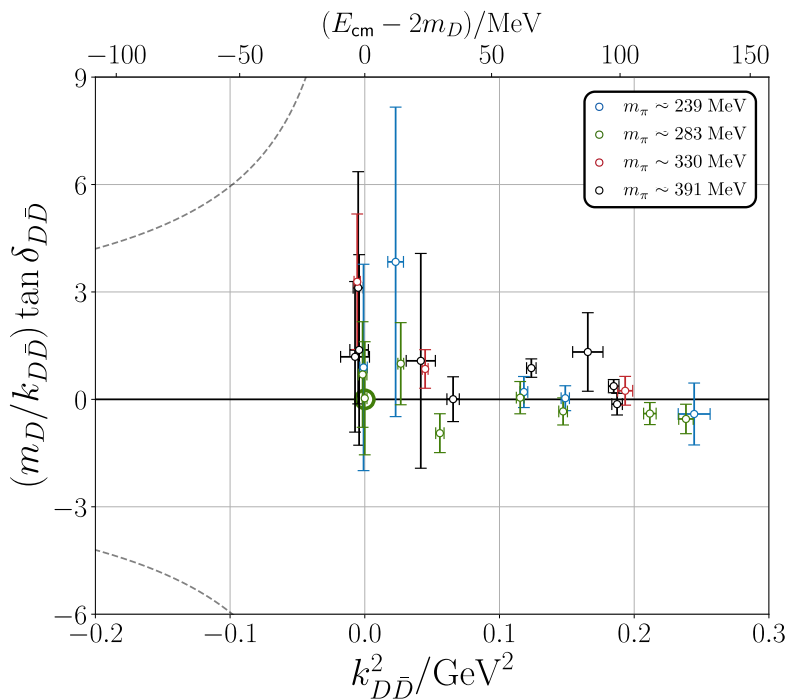
Phys.Rev.Lett. 132, 241901 (2024); Phys.Rev.D 109 (2024) 11, 114503;
arXiv:2602.09862



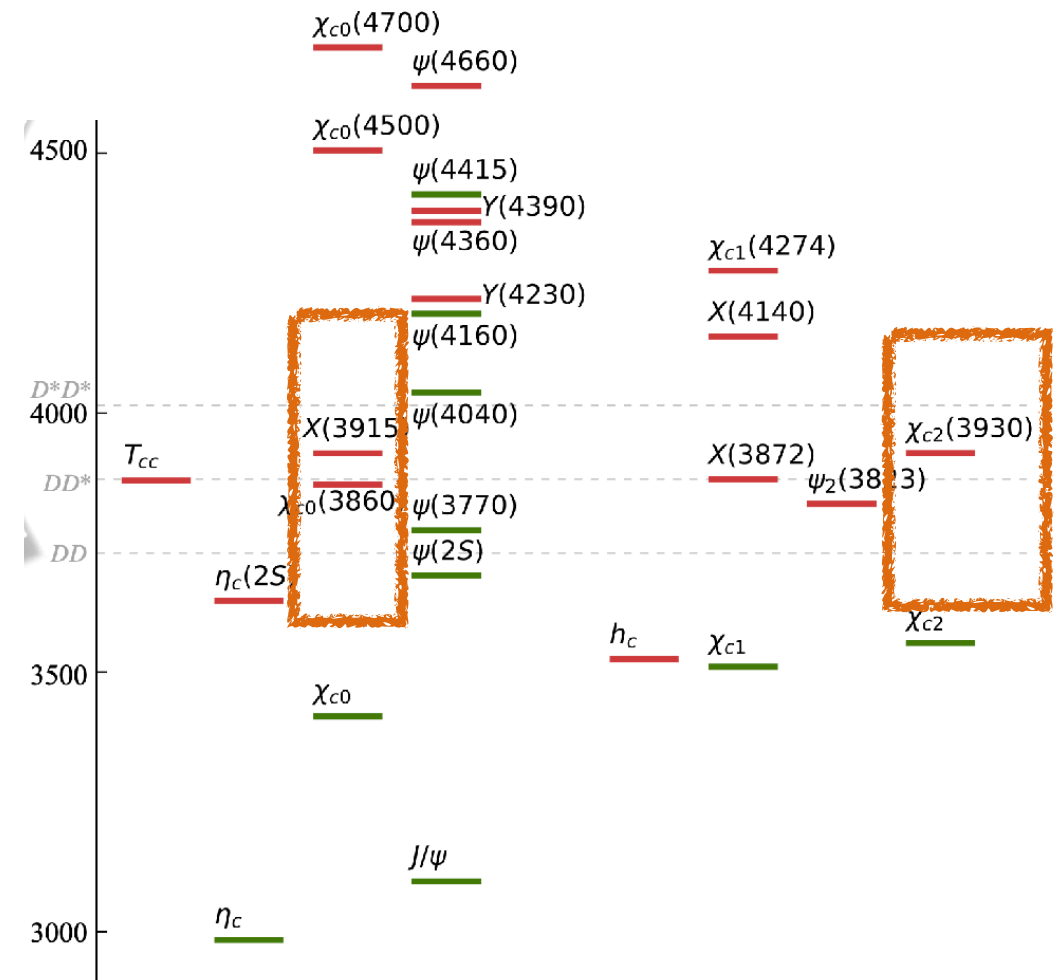
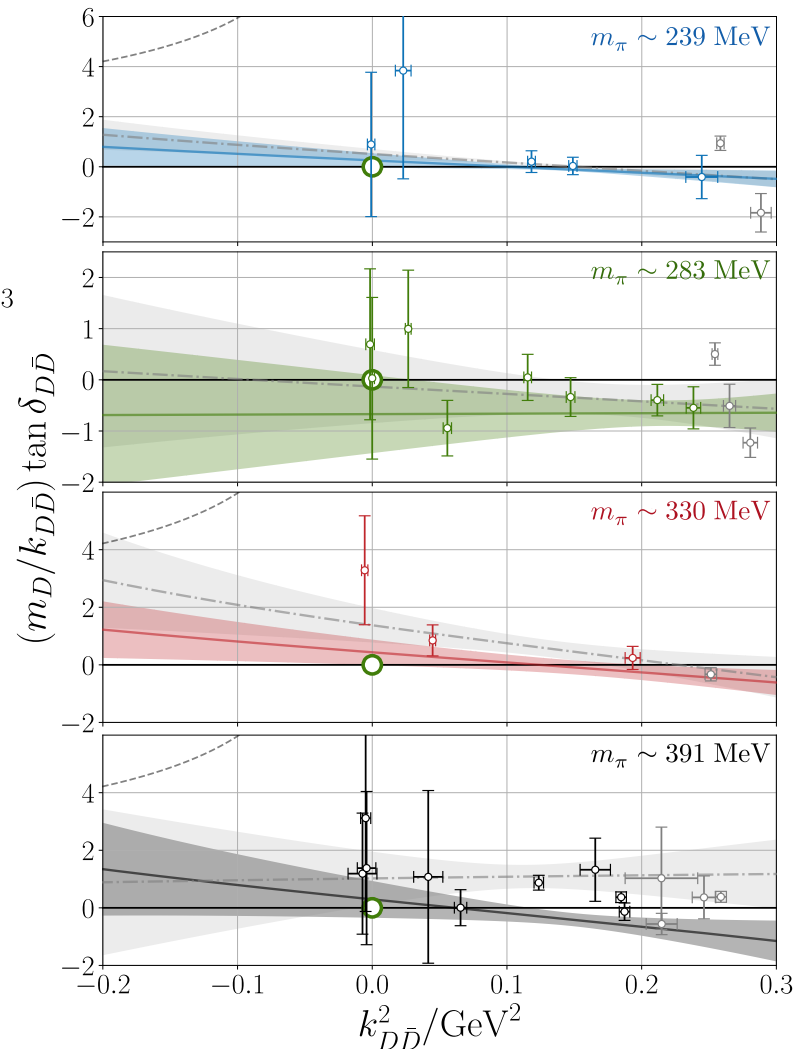
Charmonium resonances - pion mass dependence

arXiv:2602.09862

Follow-up paper - studied pion mass dependence



S-wave amplitudes -
all intersect $D\bar{D}$



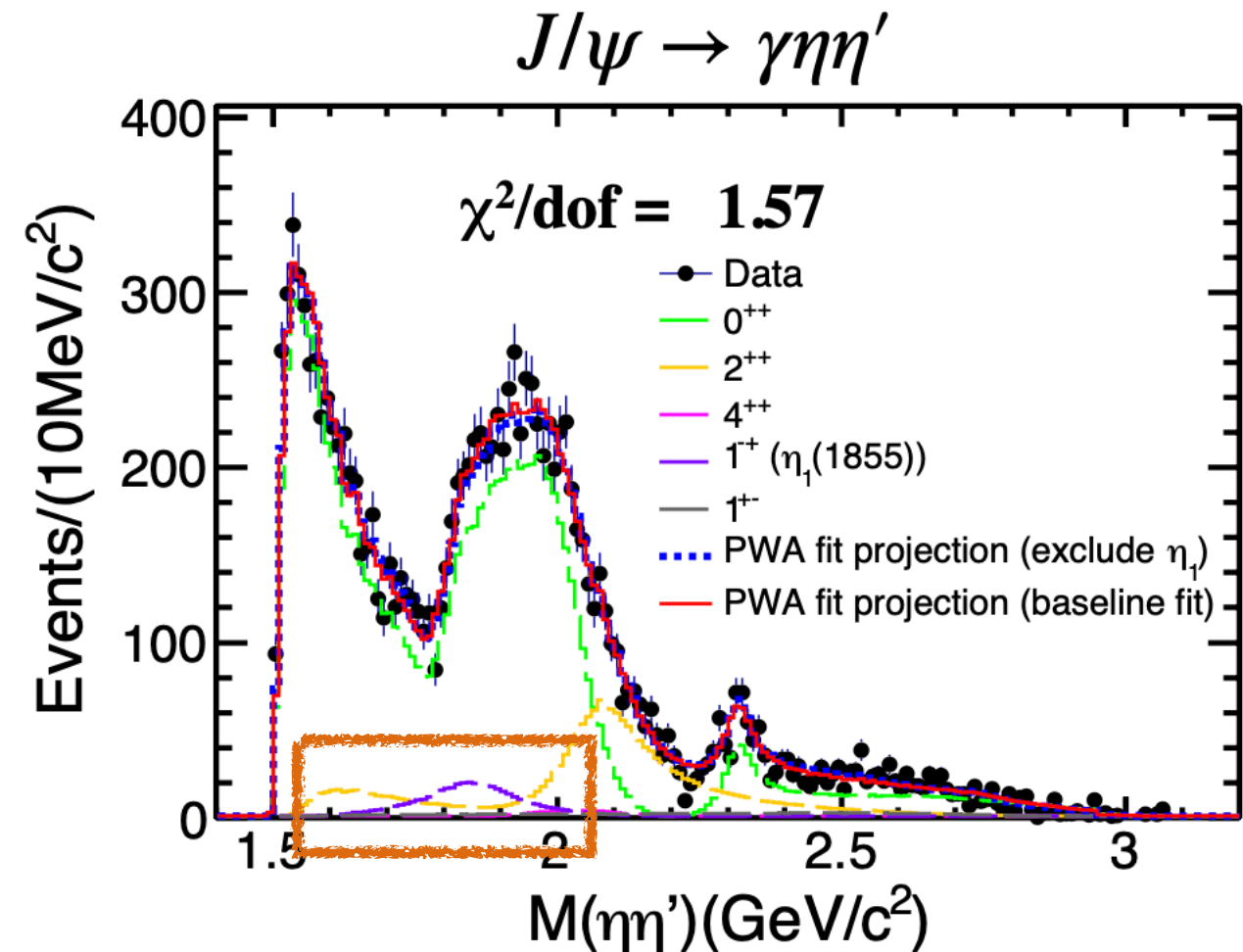
S-wave amplitudes show no signal for
virtual/bound states near $D\bar{D}$

The result suggests that production &
decay mechanisms—not simply
additional resonances—may be essential
for resolving the experimental picture.

Charmonium radiative decays - probe of structure

Transitions provide a probe of internal structure

- The electron positron collider at BESIII has produced a large amount of charmonia
 - ~10 billion J/ψ decays
 - ~3 billion $\psi(3686)$ decays
 - Clean, gluon-rich production
- Recently BESIII found an **exotic hybrid isoscalar** resonance in the radiative decay $J/\psi \rightarrow \gamma\eta\eta'$



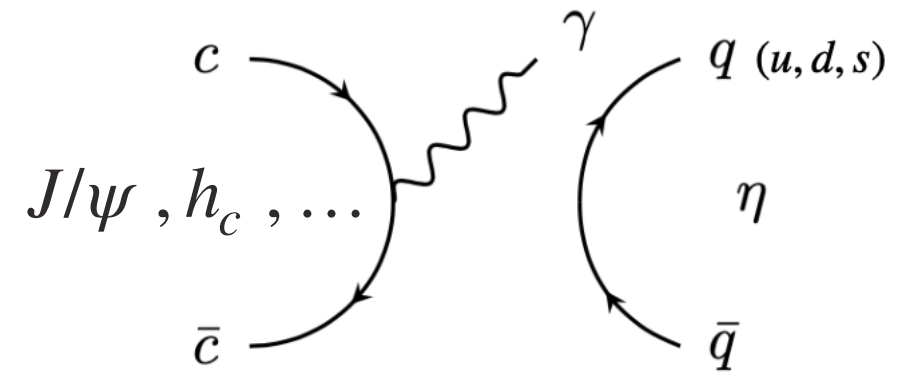
Observation of an Isoscalar Resonance with Exotic $J^{PC} = 1^{-+}$ Quantum Numbers in
 $J/\psi \rightarrow \gamma\eta\eta'$

BESIII (2022) PRL 129 192002

Charmonium radiative decays in lattice QCD

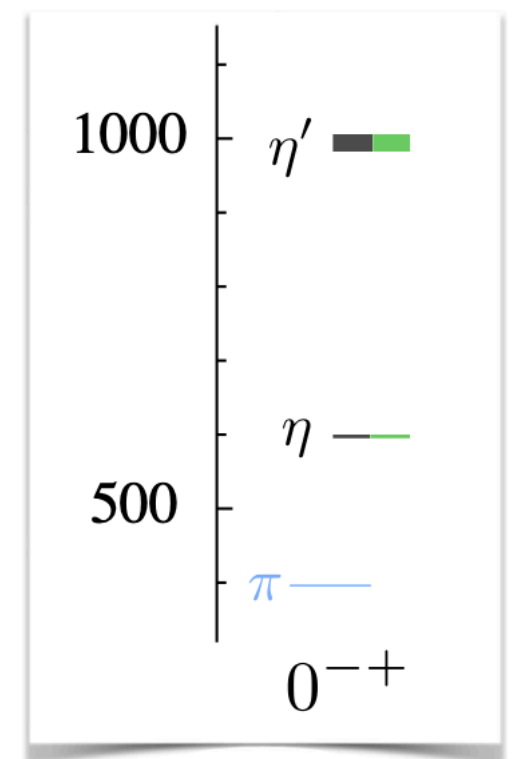
How do we calculate such decays in lattice QCD?

- three-point functions: $\langle 0 | \mathcal{O}_\eta(\delta t) \mathcal{O}_\gamma(t) [\mathcal{O}_{\mathcal{H}}(0)]^\dagger | 0 \rangle$



Challenges:

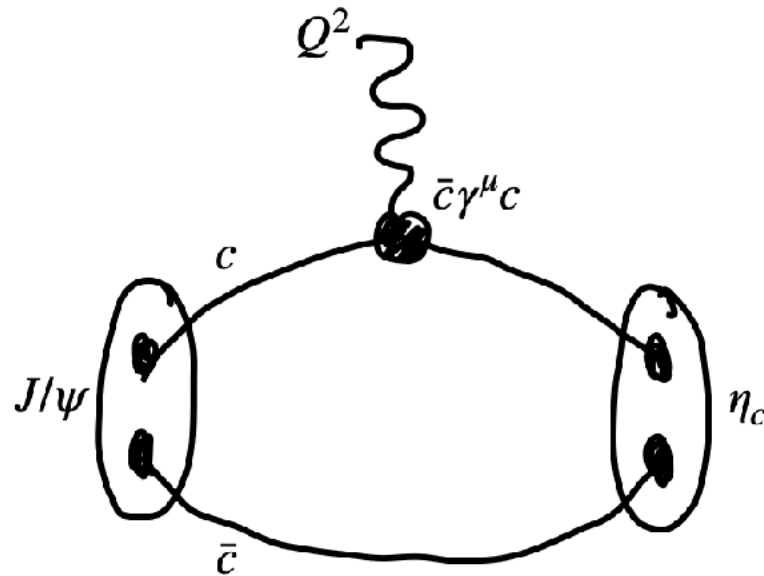
- Disconnected diagrams (noisy!)
 - Need to access both the η and η'
- Requires distinct light and strange quarks ($N_f = 2 + 1$)
- η' & h_c appear as excited states
- Large mass gap: $m_{J\psi} - m_\eta \approx 2.4$ GeV
- Need operators at large momenta for all mesons to access real-photon point $Q^2 = 0$



Rely on “*Distillation*” method (SciDAC-2 & 3)

Charmonium radiative transitions in lattice QCD

State of the art in 2006



Heavy quarks + no annihilation = clean signals

PHYSICAL REVIEW D **73**, 074507 (2006)

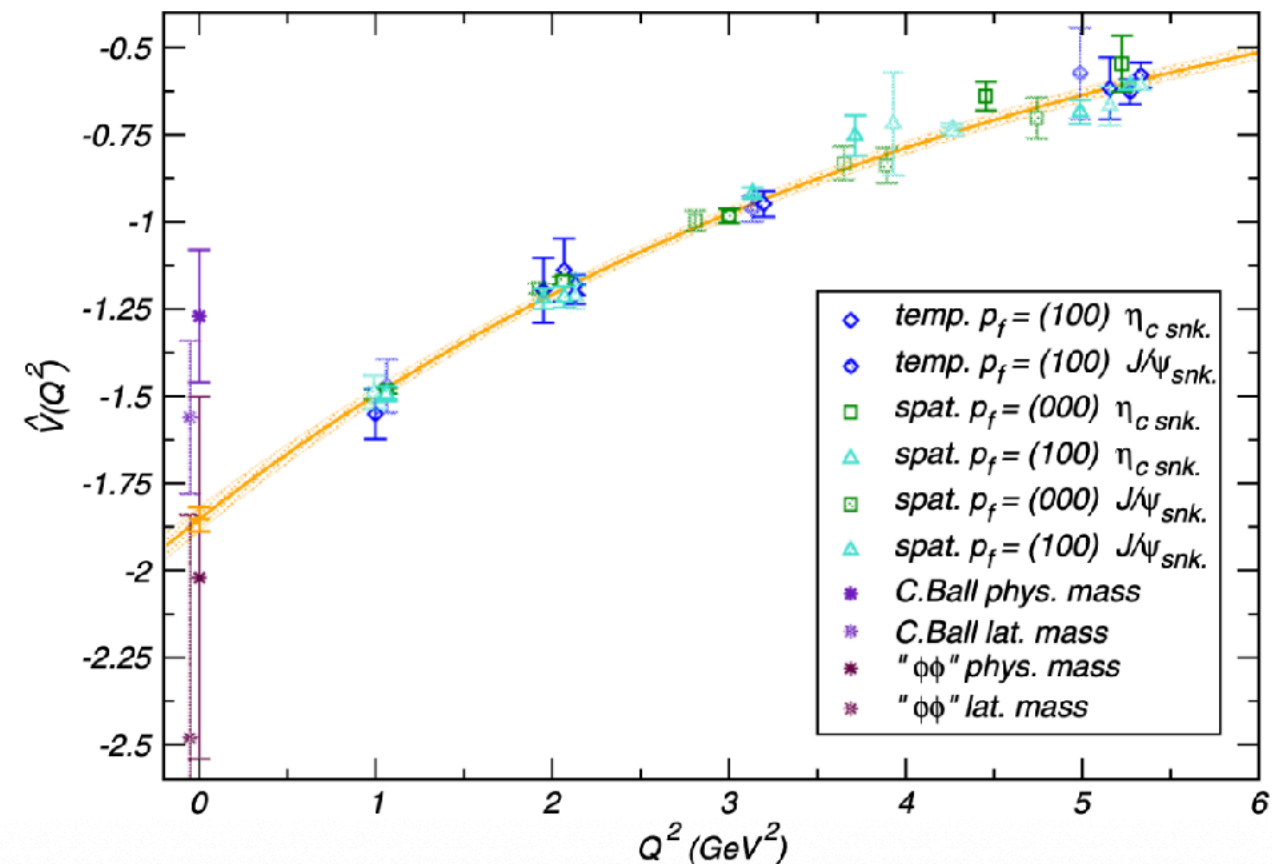
Radiative transitions in charmonium from lattice QCD

Jozef J. Dudek,* Robert G. Edwards, and David G. Richards

Jefferson Laboratory Mail-Stop 12H2, 12000 Jefferson Avenue, Newport News, Virginia 23606, USA

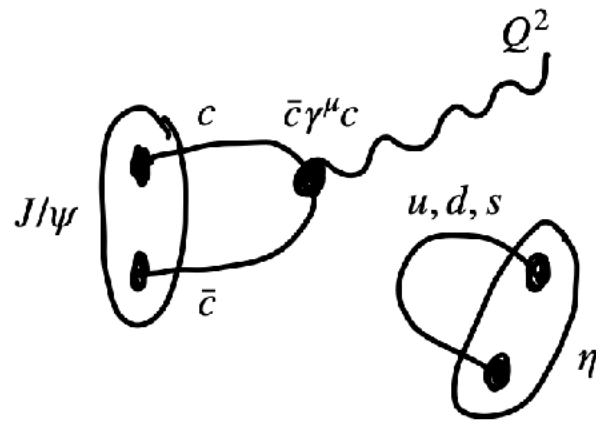
(Received 17 January 2006; published 20 April 2006)

Radiative transitions between charmonium states offer an insight into the internal structure of heavy-quark bound states within QCD. We compute, for the first time within lattice QCD, the transition form factors of various multiplicities between the lightest few charmonium states. In addition, we compute the



Charmonium **radiative decays** in lattice QCD

are challenging



Provides insight into gluonic interactions

annihilation =
noisy signals

Multigrid solvers & GPUs (SciDAC 1-5, FastMath)

Distillation & contraction methods (SciDAC 2-5, RAPIDS)

Constraints from symmetries

Wigner-Eckart theorem relates each matrix elements to a *reduced matrix element*:

$$\underbrace{\langle 0 | \Omega_{\eta(r)}^{\{\mathbf{p}'\}_{k'}^* \Lambda' \mu'} \cdot j^{\mathbf{q} \Lambda_\gamma \mu_\gamma} \cdot [\Omega_{\psi}^{\{\mathbf{p}\}_k^* \Lambda \mu}]^\dagger | 0 \rangle}_{\text{Combinations of all}} = \underbrace{\mathcal{C}(\{\mathbf{p}\}_k^* \Lambda \mu; \{\mathbf{p}'\}_{k'}^* \Lambda' \mu'; \mathbf{q} \Lambda_\gamma \mu_\gamma)}_{\text{Clebsch-Gordan coeff.}} \cdot \underbrace{\mathbb{C}(\{\mathbf{p}'\}^* \Lambda'; \mathbf{q} \Lambda_\gamma; \{\mathbf{p}\}^* \Lambda)}_{\text{reduced correlator}}$$

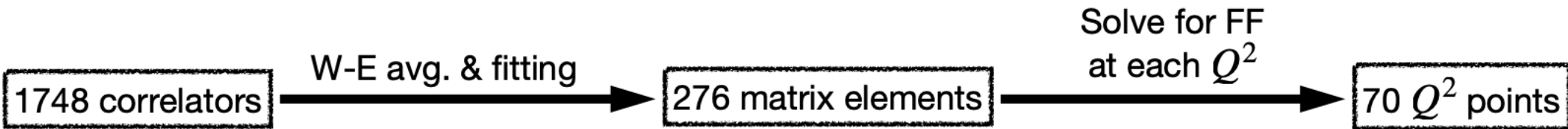
- allowed lattice rotations of the momenta
- rows of the irreps

Provides additional constraints - well suited to Distillation

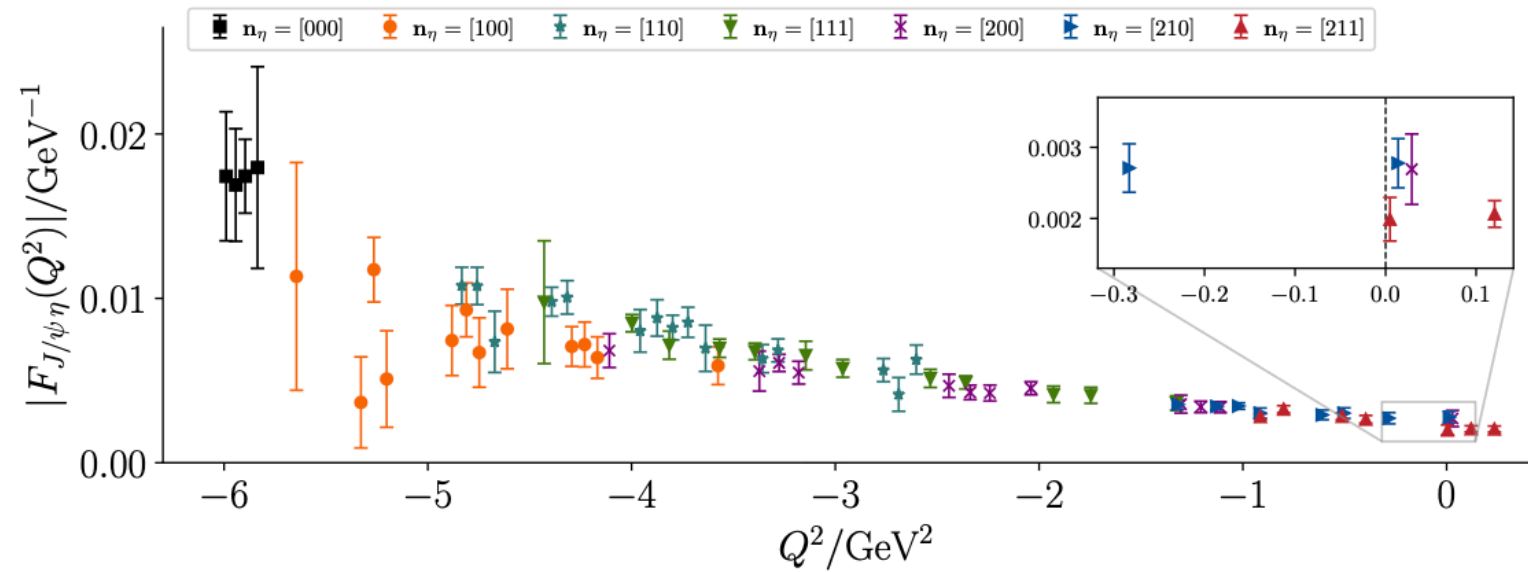
SciDAC-5

Techniques broadly across hadron physics...

$J/\psi \rightarrow \gamma \eta^{(\prime)}$ radiative decays

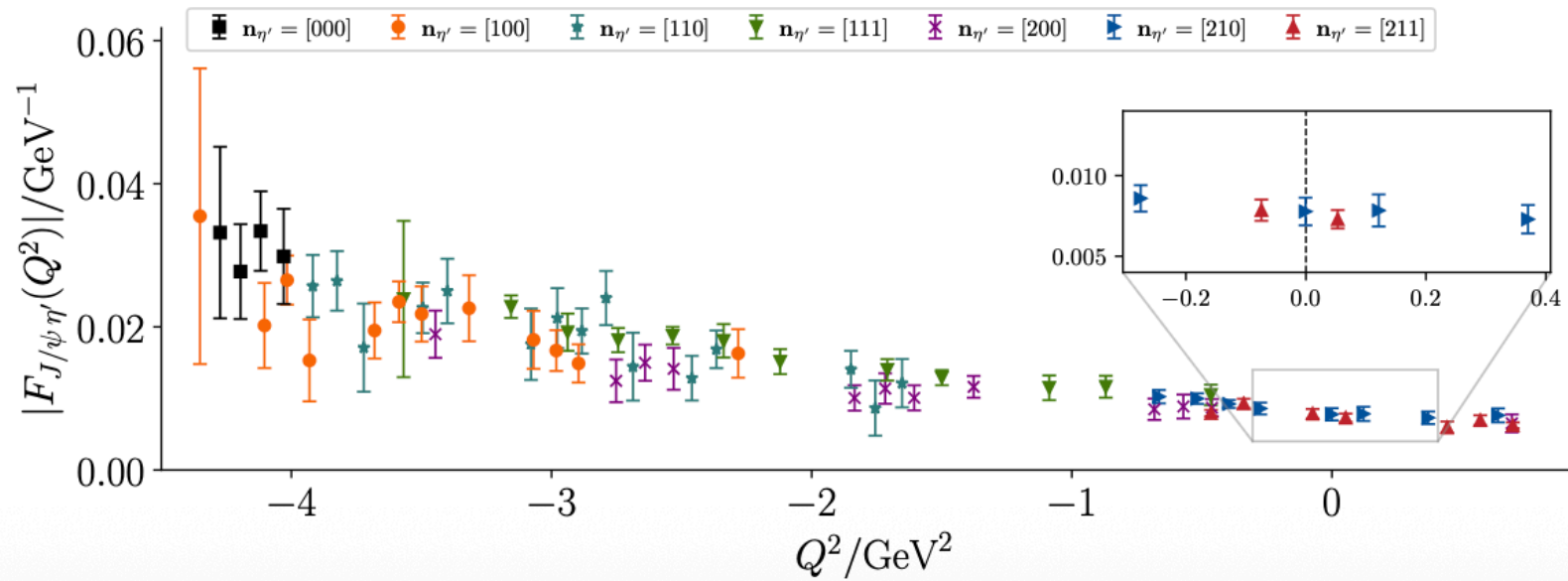


$J/\psi \rightarrow \gamma \eta$



High-momentum operators are essential to reach $Q^2 = 0$

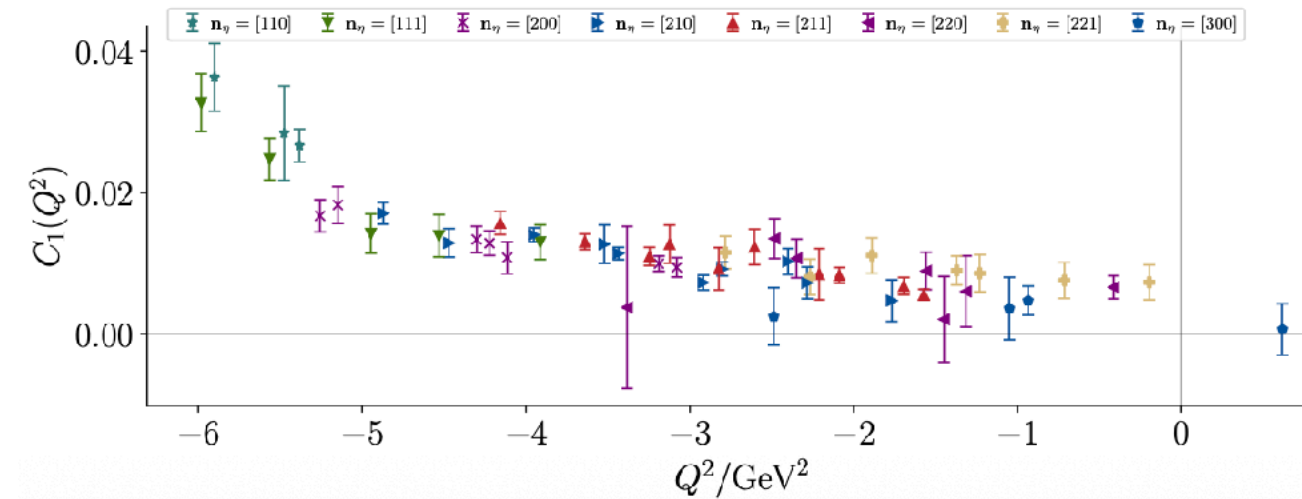
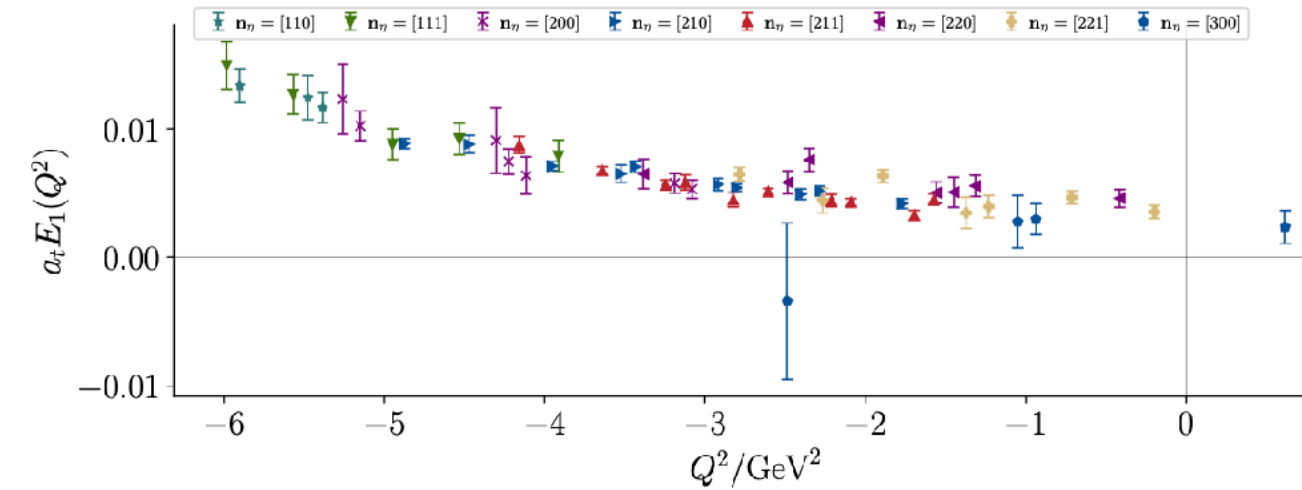
$J/\psi \rightarrow \gamma \eta'$



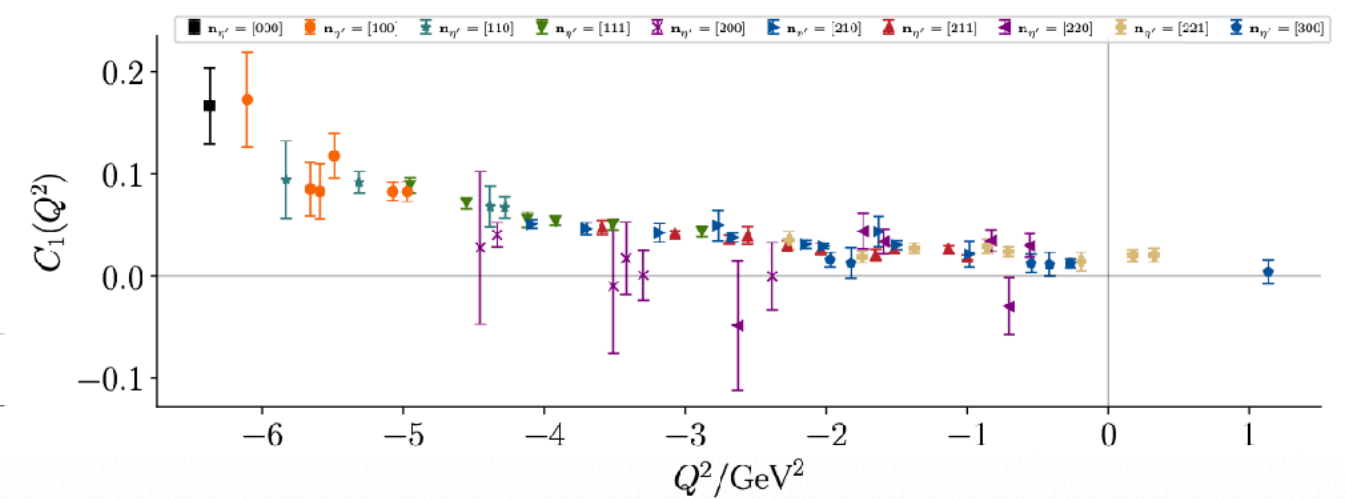
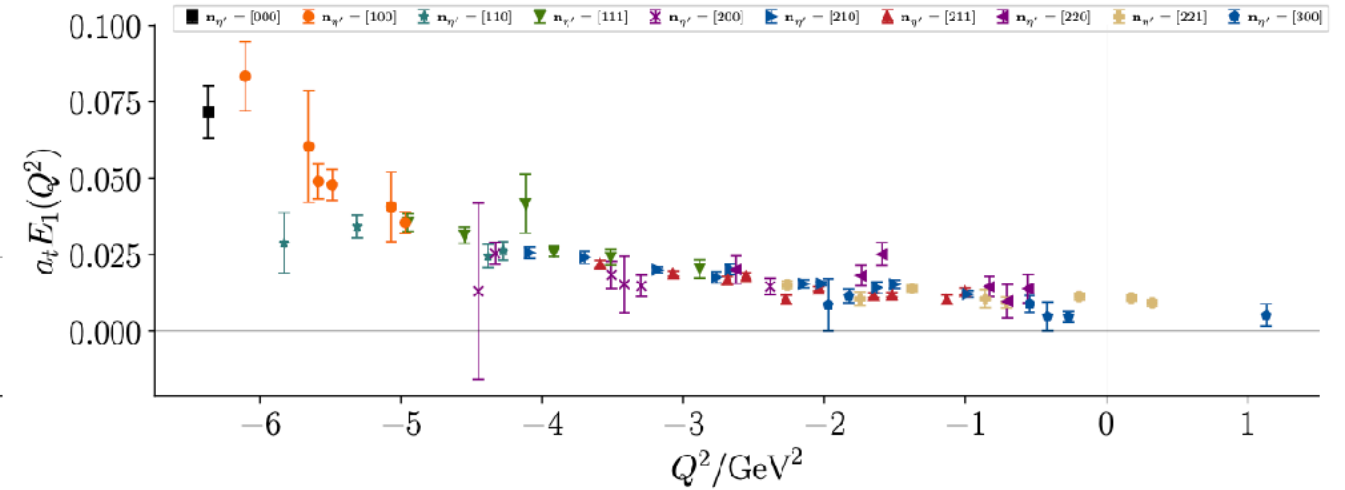
$h_c \rightarrow \gamma \eta^{(\prime)}$ radiative decays

Has an electric dipole $E_1(Q^2)$ and longitudinal $C_1(Q^2)$ form-factor

$$h_c \rightarrow \gamma \eta$$

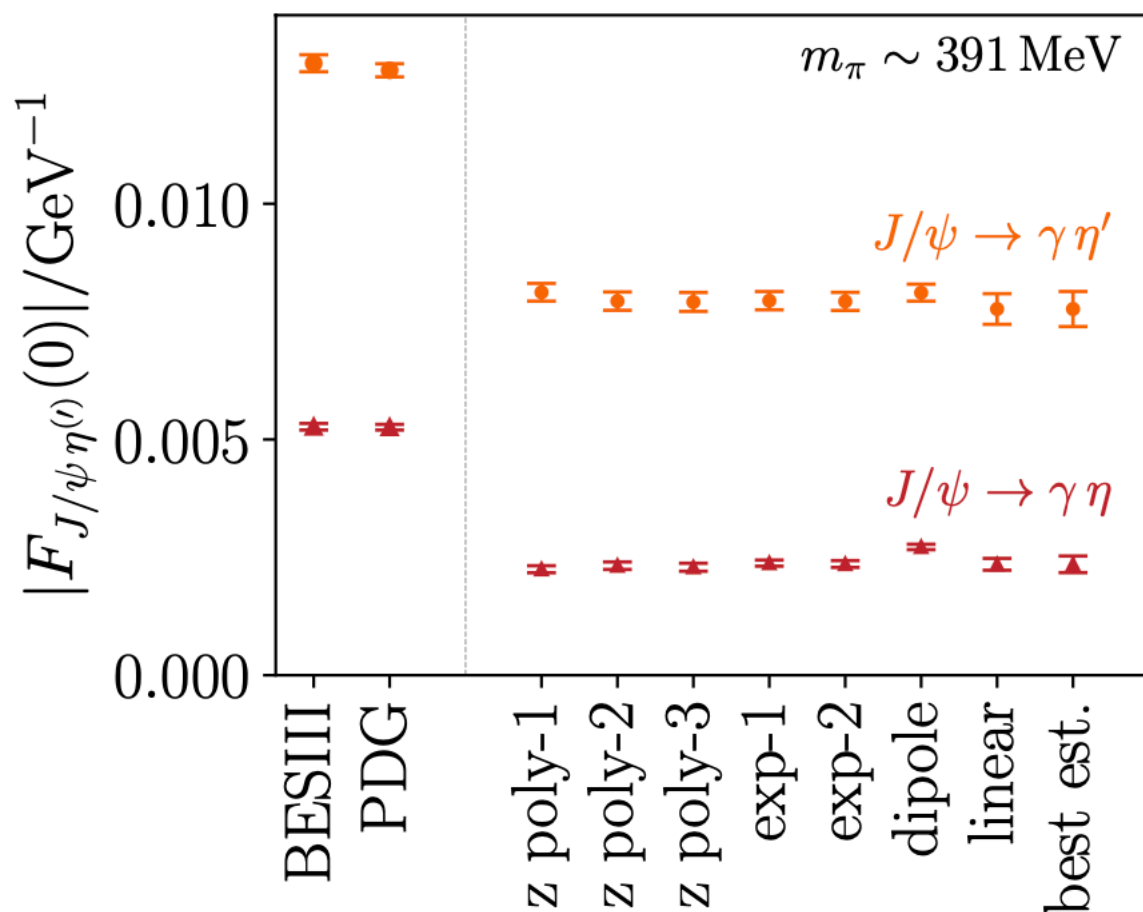


$$h_c \rightarrow \gamma \eta'$$



Comparison with BESIII results

$$J/\psi \rightarrow \gamma \eta^{(\prime)}$$



Discrepancy with experiment

vacuum topology? light quark mass? $U(1)_A$ anomaly?

Future - lower pion masses

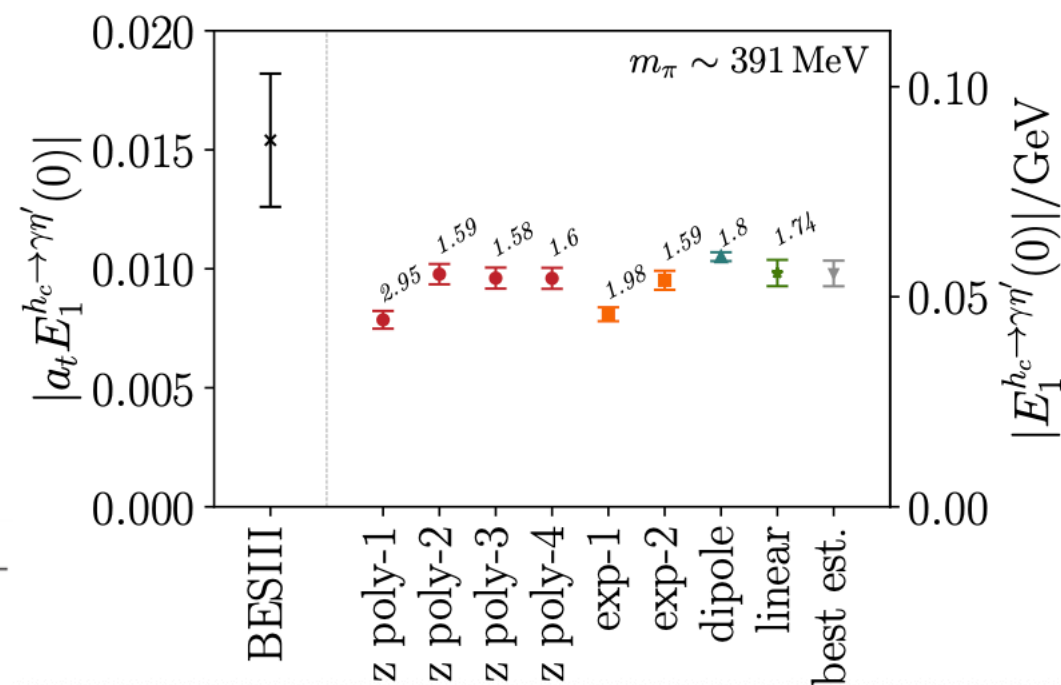
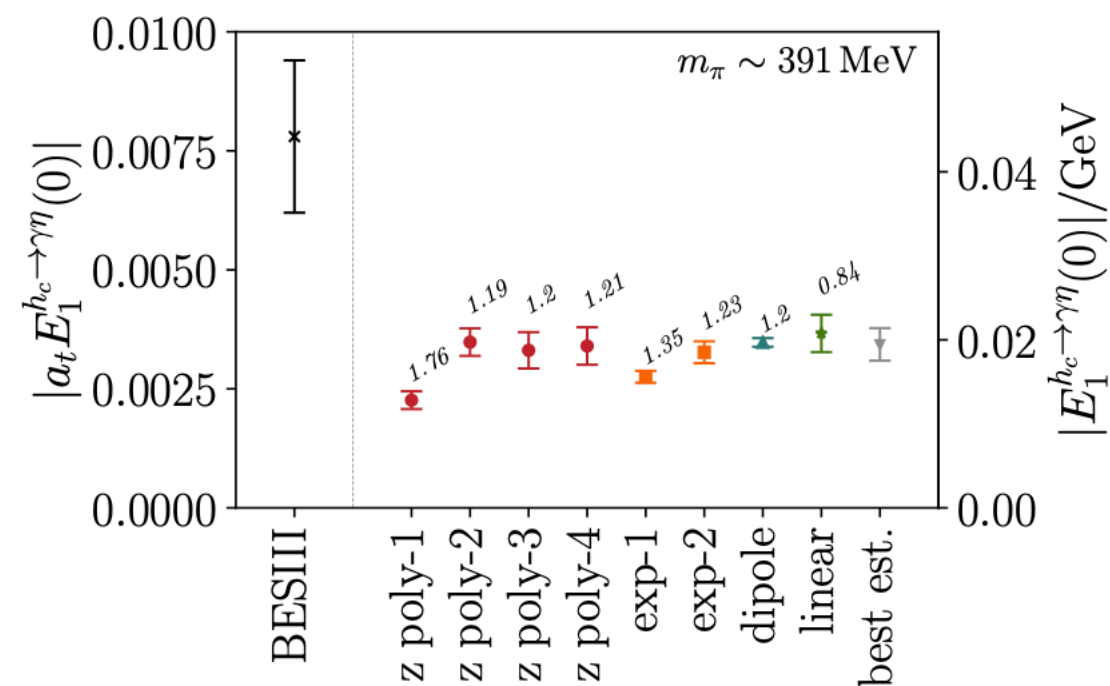
PHYSICAL REVIEW LETTERS 135, 161904 (2025)

η and η' Production in J/ψ Radiative Decays from Quantum Chromodynamics

PHYSICAL REVIEW D 112, 074505 (2025)

η and η' meson production in J/ψ radiative decays from lattice QCD

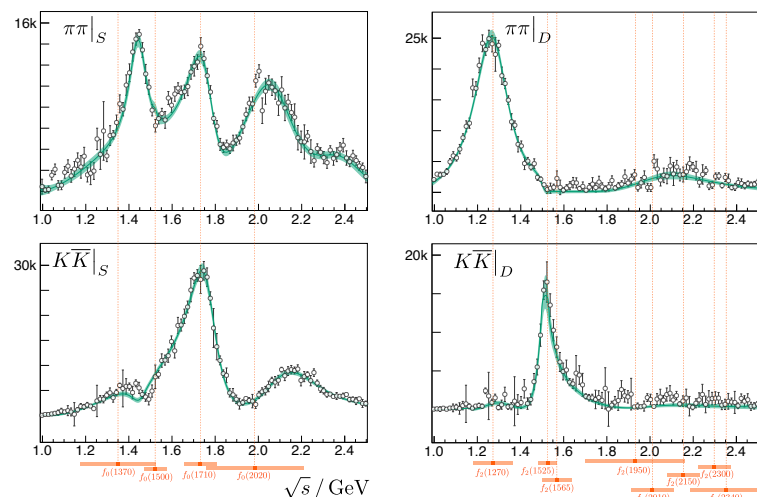
$$h_c \rightarrow \gamma \eta^{(\prime)}$$



Extensions of the technology

Compute scalar and tensor resonance production

$$J/\psi \rightarrow \gamma (\pi\pi, K\bar{K}, \dots)$$



f_0, f_2 resonances previously studied in coupled-channel scattering using LQCD

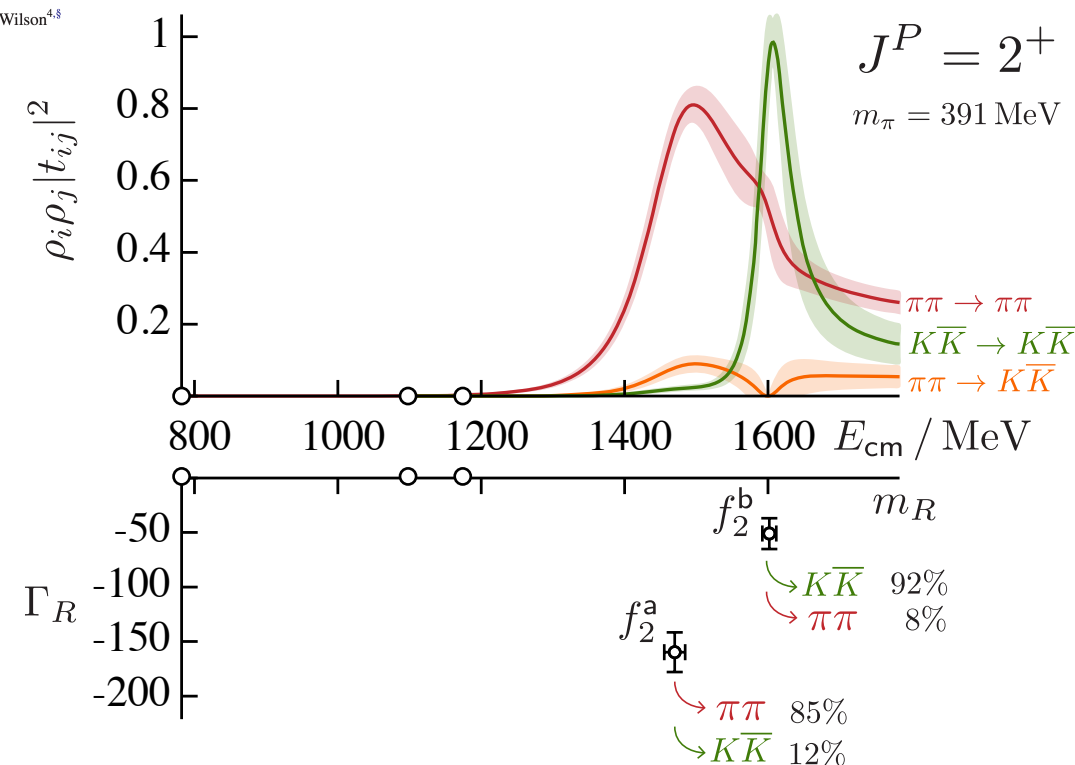
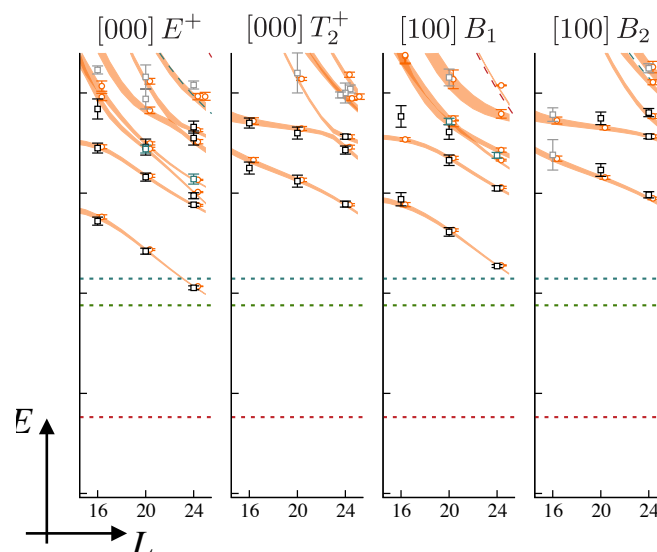
PHYSICAL REVIEW D **97**, 054513 (2018)

Editors' Suggestion

Isoscalar $\pi\pi, K\bar{K}, \eta\eta$ scattering and the σ, f_0, f_2 mesons from QCD

Raul A. Briceño,^{1,2,*} Jozef J. Dudek,^{1,3,†} Robert G. Edwards,^{1,‡} and David J. Wilson^{4,§}

(for the Hadron Spectrum Collaboration)

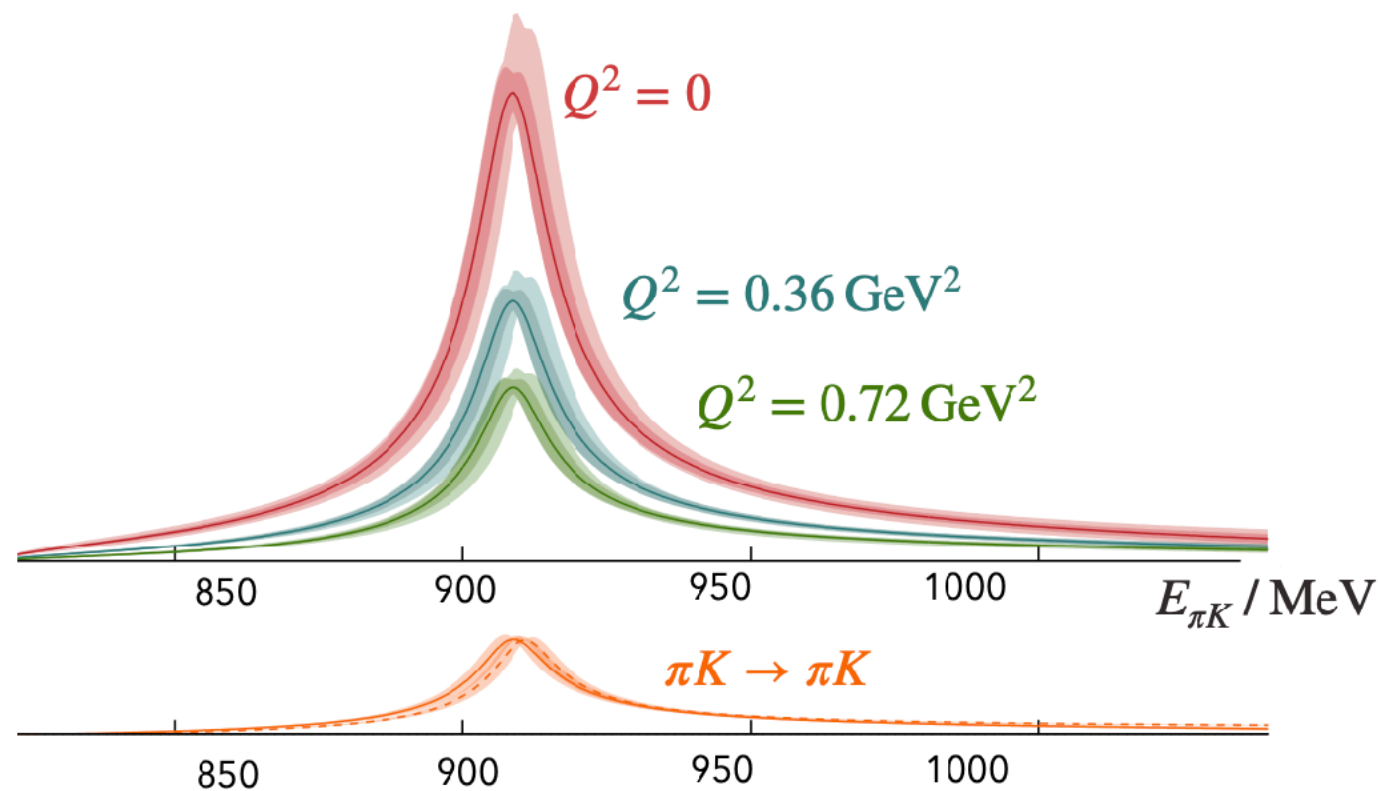
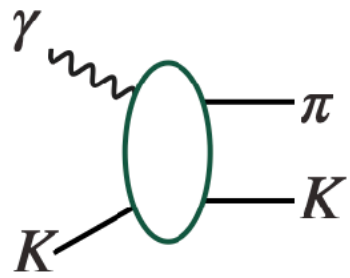


Build up systematically toward computation of hybrid/exotic $J/\psi \rightarrow \gamma \eta \eta'$. $J^{PC} = 1^{-+}$

Goal: reveal gluonic structure

Coupling resonances to currents

$$\gamma K \rightarrow \pi K$$

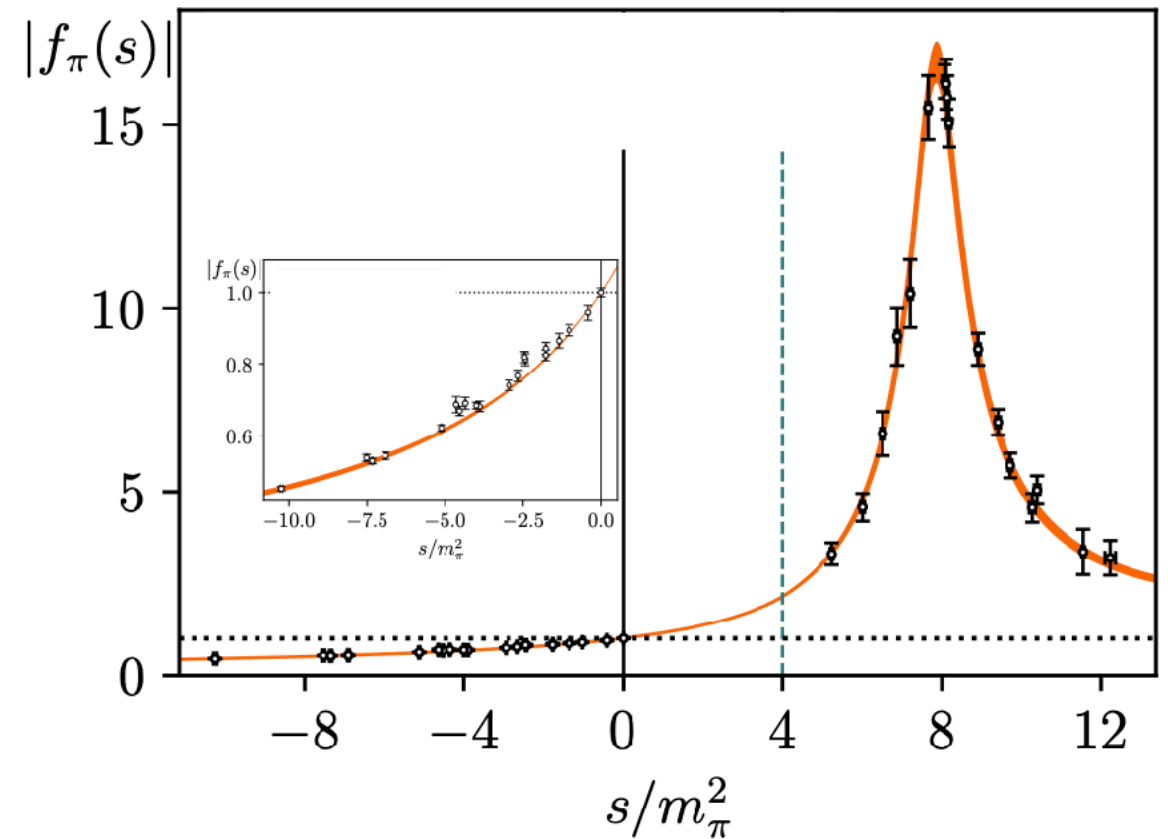


$J^P = 1^-$ K^* resonance appears

$$m_\pi \sim 283 \text{ MeV}$$

PRD 106, 114513 (2022)

$$e^+e^- \rightarrow \pi\pi \text{ \& } e^-\pi \rightarrow e^-\pi$$



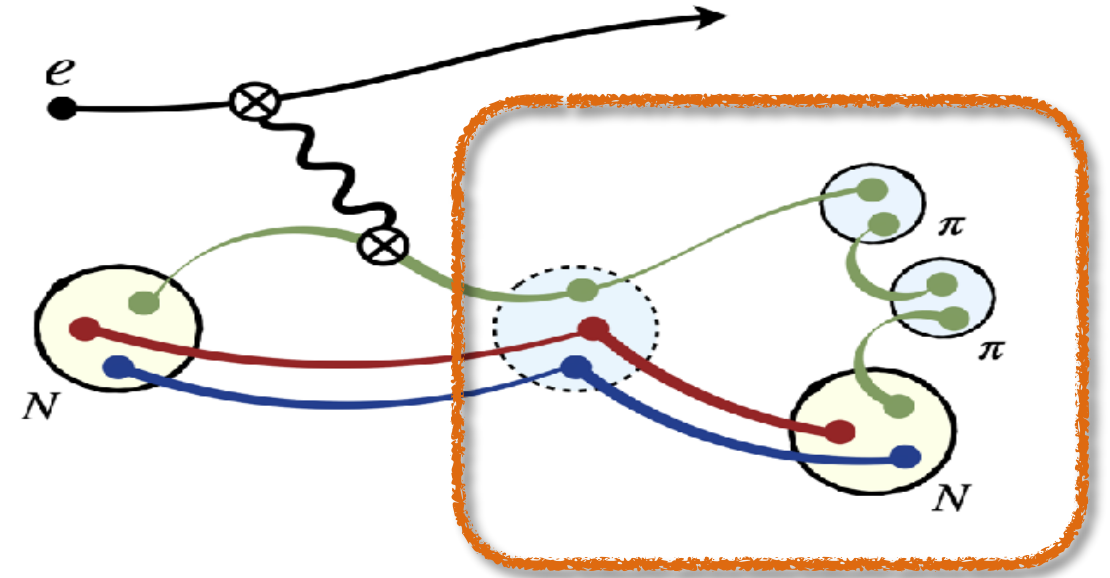
$J^P = 1^-$ ρ resonance appears

$$m_\pi \sim 283 \text{ MeV}$$

PRD 110, 094505 (2024)

3-body decays

- Building to cases like nucleons
- 3-body formalism



3-body - an old Problem!



@三体电影

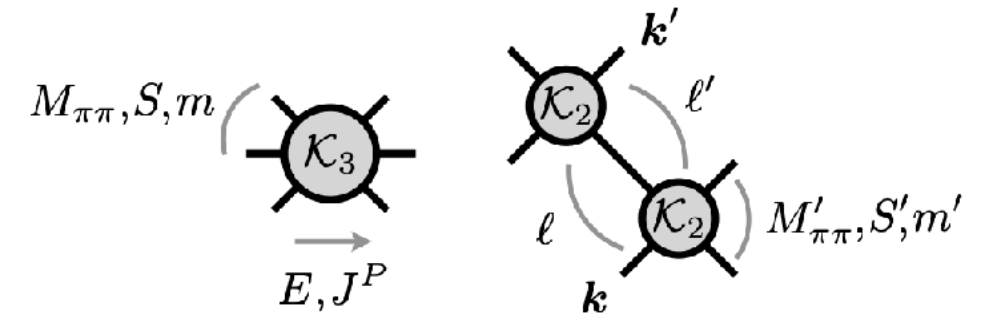
Progress with 3-body scattering

- **Simplified 3-body formalism**

- Finite-V energies used to constrain 3-body (& 2-body) K-matrix parameterizations
- Integral equations relate to physical amplitudes

- **Isospin= 3 $\pi\pi\pi$**

Two contributions to amplitudes



Here, $\ell = S = J = 0$

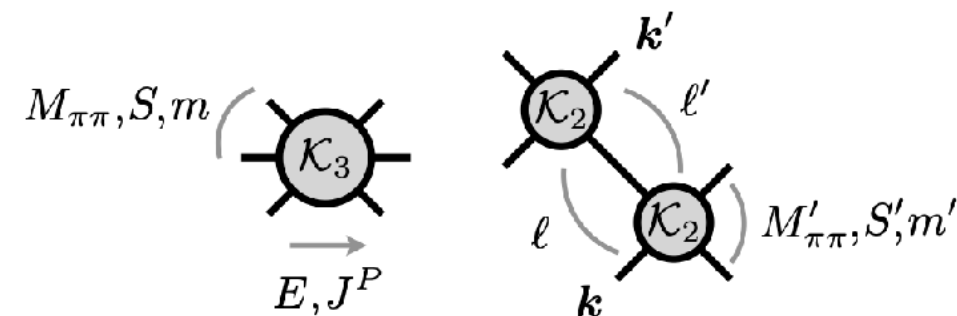
Progress with 3-body scattering

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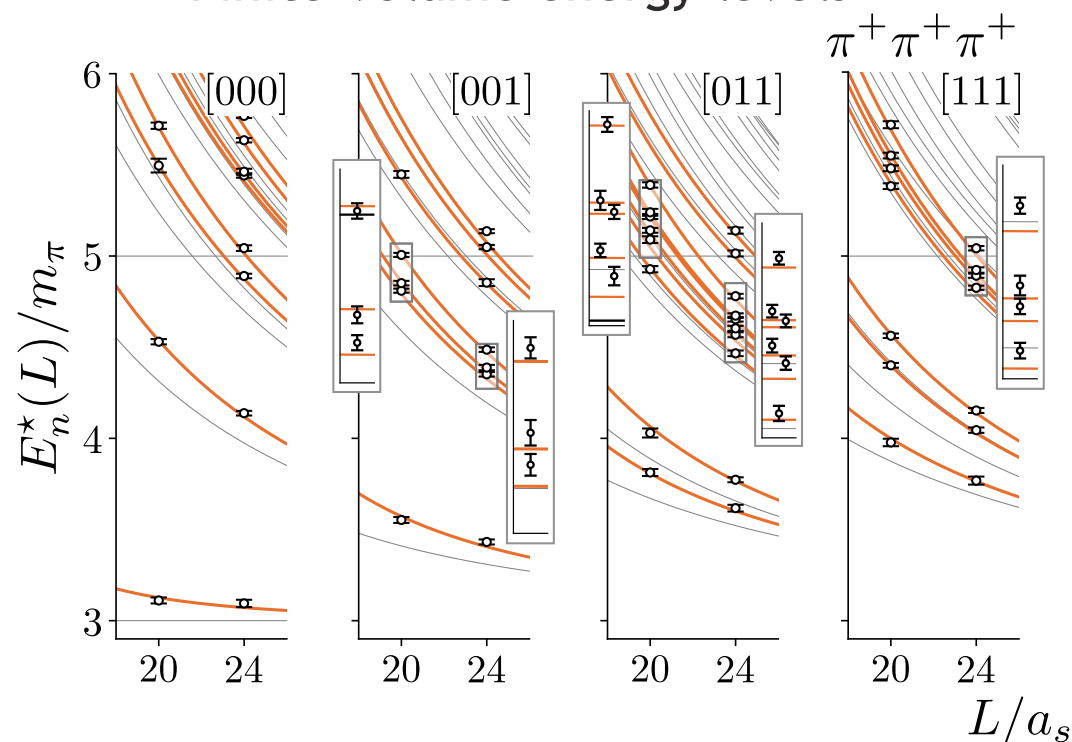
- Isospin= 3 $\pi\pi\pi$**

Two contributions to amplitudes

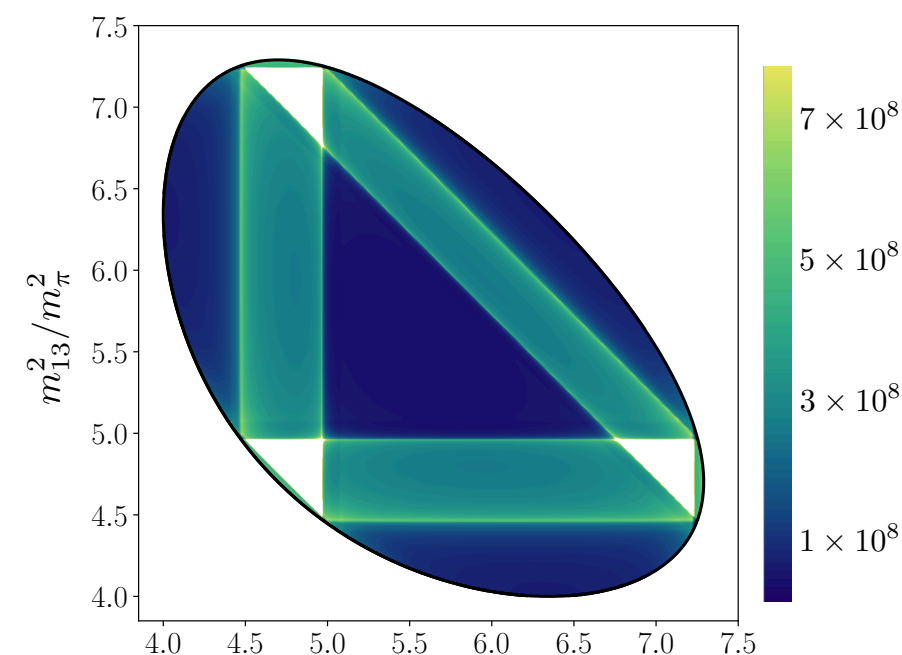


Here, $\ell = S = J = 0$

Finite-volume energy levels



3-body amplitudes (Dalitz plot)



HADRON SPECTRUM: PRL 123
EDITOR'S SUGGESTION

3-body with 2-body intermediate resonance

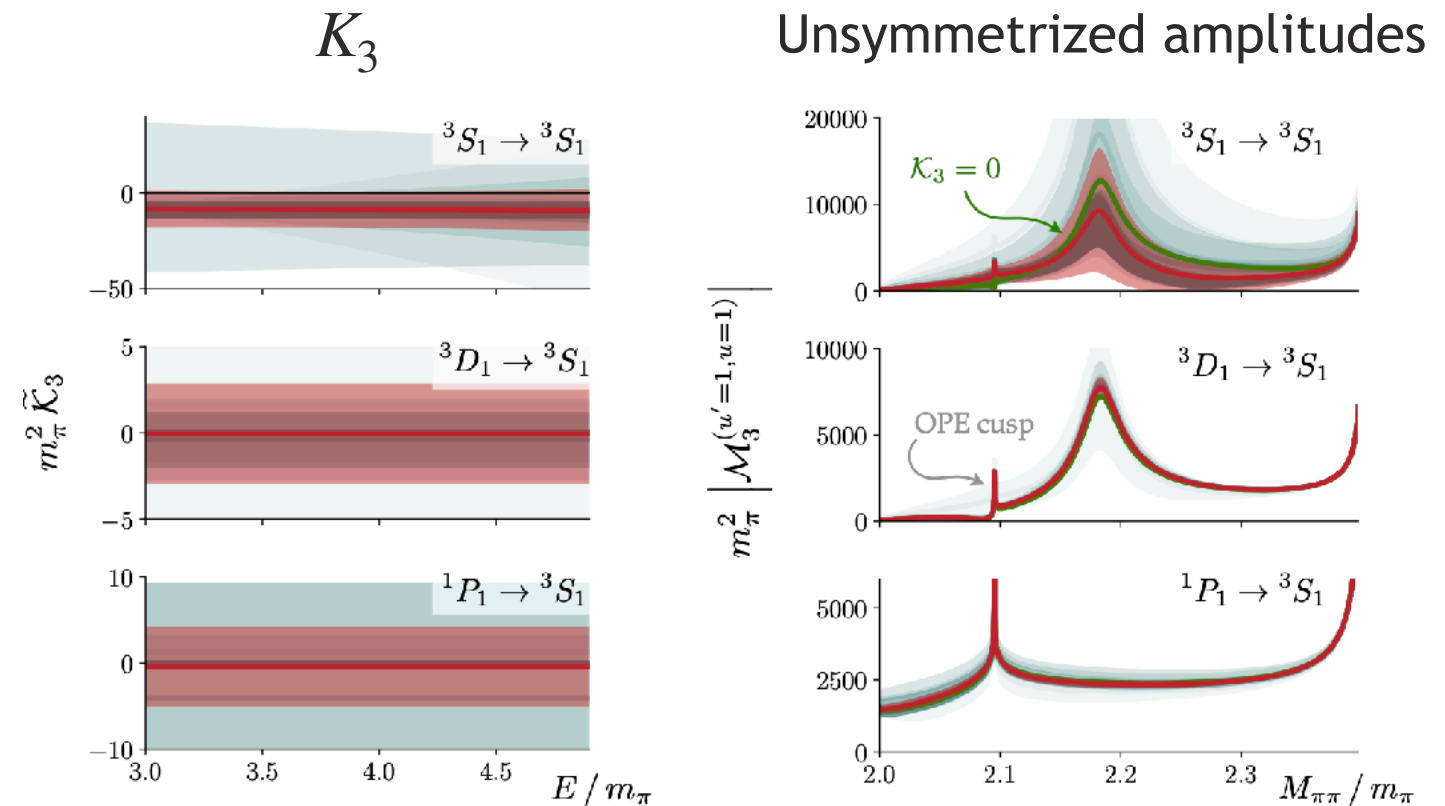
Accepted in PRL (arXiv:2510.24894)

- Development of full 3-body formalism

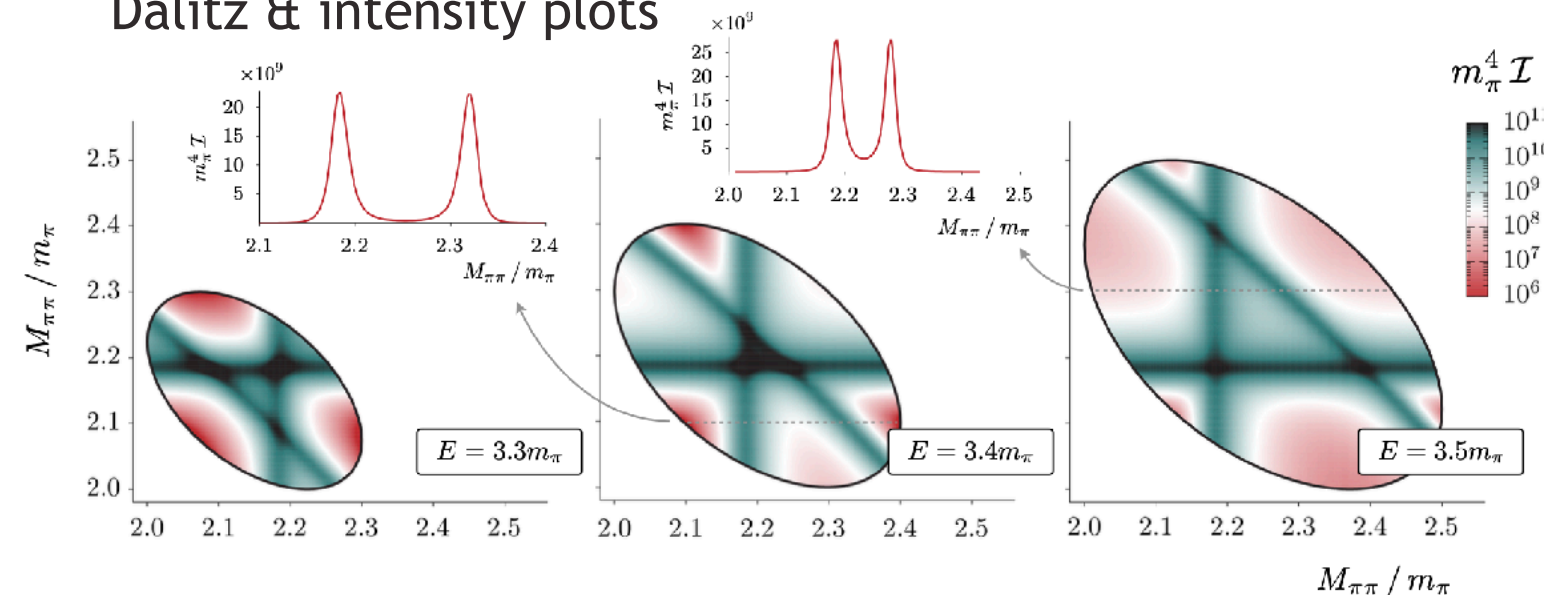
- Lattice Finite-V energies used to constrain 3-body (& 2-body) K-matrix parameterizations
- Integral equations used to relate to physical amplitudes

- Isospin= 2 $\rho\pi$ and $\pi\pi\pi$

- Full/general formalism including intermediate subsystems

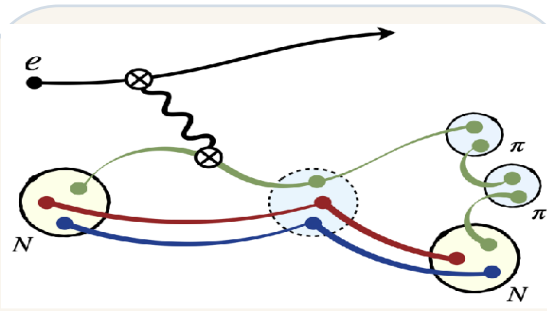


Dalitz & intensity plots



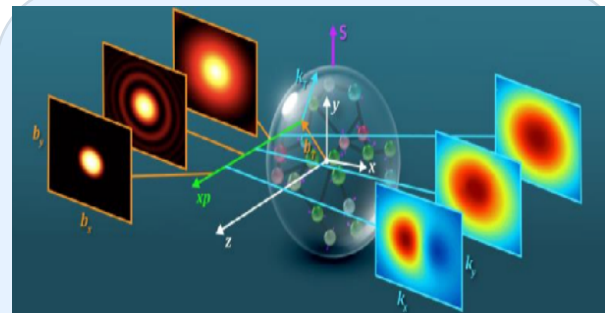
Future: I=0 & 1 heavy/light quark systems

Spectroscopy underpins structure & discovery



Hadron spectroscopy

Resonance spectrum & dynamics



3D femtoscale imaging

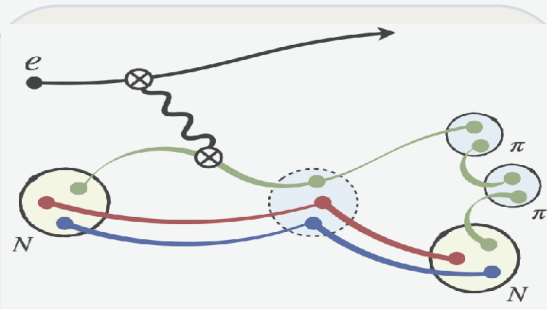
Quark & gluon structure



Nuclear structure

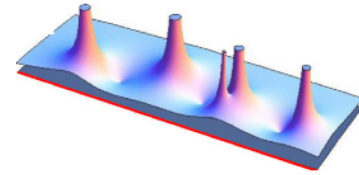
Precision nuclear & neutrino observables

Spectroscopy underpins structure & discovery



Hadron spectroscopy

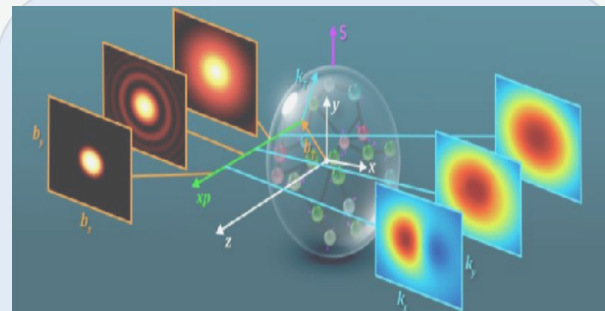
Resonance spectrum & dynamics



Resonances are essential inputs, not merely final states

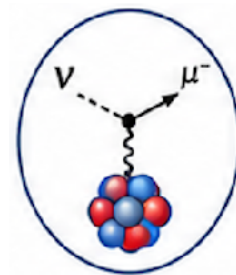
First-principles - determine masses, widths, couplings, & transition amplitudes needed to interpret scattering and decay experiments

Resonance properties feed directly into calculations of hadron structure & nuclear response, reduce model dependence in searches for new physics.



3D femtoscale imaging

Quark & gluon structure



Example: neutrino interactions

For DUNE, axial form factors of the nucleon & transition form factors with Delta resonance required to describe neutrino-nucleus reactions

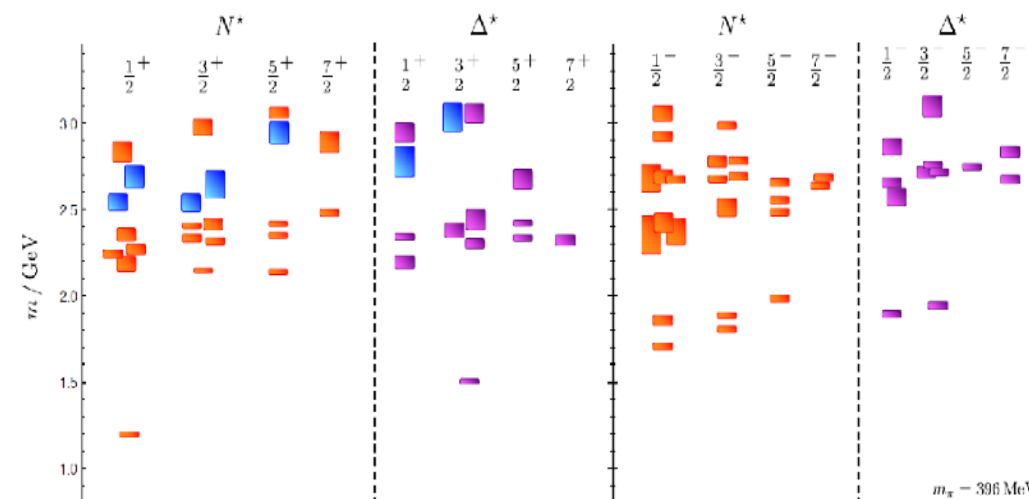
Without controlled resonance information, uncertainties in strong-interaction physics can obscure the extraction of neutrino properties.



Nuclear structure

Precision nuclear & neutrino observables

SciDAC-2: first glimpses of the excited baryon spectrum



m

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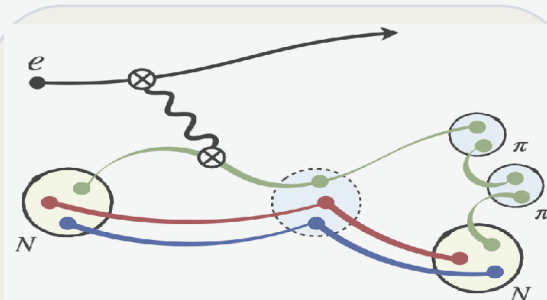
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$\langle N^* | J | N \rangle$

SciDAC-5:
resonance
masses, widths,
couplings, and
transition form
factors

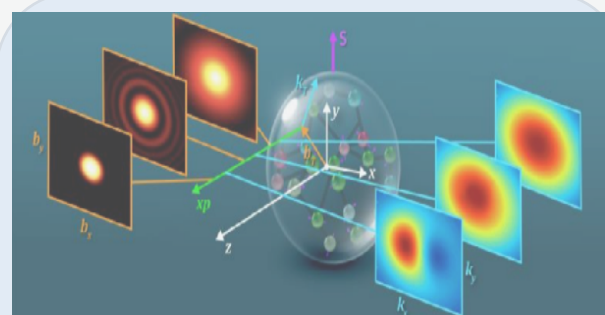
Excited state baryon spectroscopy from lattice QCD
PRD 83 074508 (2011)

Hadron structure underpins the EIC & HL-LHC



Hadron spectroscopy

Resonance spectrum & dynamics



3D femtoscale imaging

Quark & gluon structure



Nuclear structure

Precision nuclear & neutrino observables

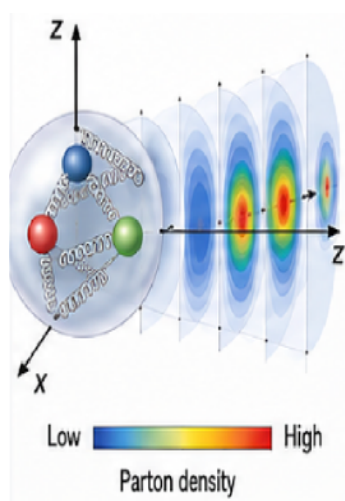


Parton structure is an essential input to discovery science

First-principles calculations of hadron structure determine partonic content of proton - provides theory constraints on PDFs & related observables.

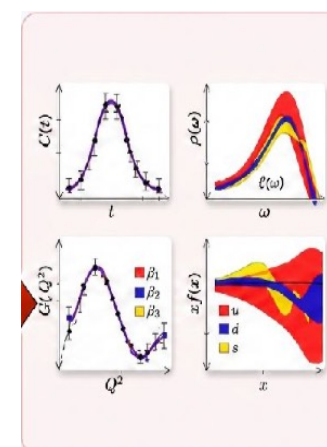
Required to interpret precision measurements at both the EIC & HL-LHC.

EIC: revealing the proton in 3D



- PDFs/TMDs/GPDs map quark & gluon mom., spin, and spatial structure
- LQCD supplies benchmark moments, matrix elements, & nonperturbative constraints
- Precision theory sharpens the interpretation of EIC measurements

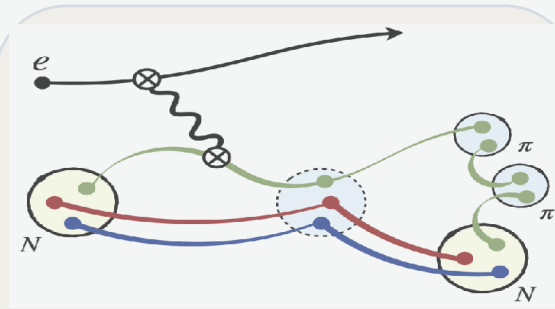
HL-LHC: precision tests



- PDFs are a dominant theory uncertainty for key benchmark processes
- For some flagship observables, PDF uncertainties account for ~90% of the theory uncertainty budget
- Better hadstruct strengthens SM tests & new-physics sensitivity

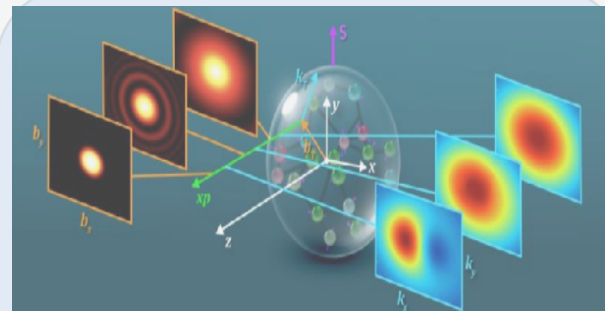
SciDAC-5 provides the algorithms, software, and workflows that make precision hadron-structure calculations possible at scale.

Nuclear structure underpins the EIC & DUNE



Hadron spectroscopy

Resonance spectrum & dynamics



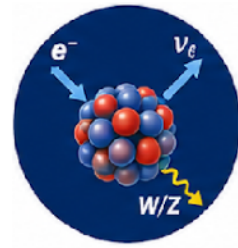
3D femtoscale imaging

Quark & gluon structure



Nuclear structure

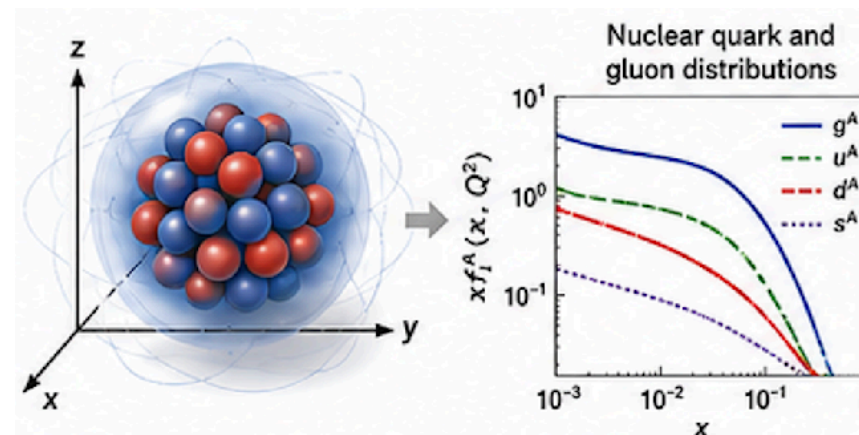
Precision nuclear & neutrino observables



Precision nuclear structure essential to EIC & DUNE

First-principles calculations of nuclei determine nuclear structure, response functions, quark & gluon distributions in nuclei, & electroweak matrix elements needed to interpret precision experiments & constrain new physics.

Nuclear structure: from nuclei to quark-gluon distribution



- Precision many-body calculations reveal nuclear structure & response
- Nuclear quark & gluon distributions connect nuclei to QCD
- Support interpretation of scattering & electroweak observables

SciDAC-enabled algorithms, software, and workflows are building the capability for precision nuclear theory at scale.

SciDAC-5 is a partnership

- Strong partnerships with FastMath & Rapids, and ASCR/NP supported community
- SciDAC a huge boost for LQCD & NP
 - present science was not possible 10 years ago
 - fresh perspective on entire program
- Is significantly impacting our calculations
 - huge advance in gauge generation - accelerated our science campaigns
 - accelerating analysis campaigns on leadership & local resources
 - better tools for (more easily) improving performance

